

MapMet project

Tumor Cell Cluster Reliability and BM vs PT Enrichment Analysis

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Contents

IMPORTANT	2
Overview	2
1. Setup and Data Loading	3
1.1 Libraries and parameters	3
1.2 Metadata preparation	4
1.3 DE cohort overview	5
2.1 Rationale	9
2.2 Aggregation unit definition	9
2.3 Bootstrap resampling	10
2.4 Instability scores and flagging	10
2.5 Instability plot — BM+DE units highlighted	12
All units	12
BM+DE bootstrap ribbons	13
BM+DE CI width summary	16
3. Artifact Detection — Three-Test Framework	18
3.1 Test 1 — Per-cluster bootstrap instability	18
3.2 Test 2 — Detection rate analysis	20
3.3 Test 3 — Bootstrap CI overlap with zero	23
3.4 Combined artifact verdict	25
4. Validation of Flagged Clusters in Primary Tumor	28
4.1 PT detection rates for flagged clusters	28
4.2 BM vs PT detection rate comparison	30

5. BM vs PT Enrichment — Testing the Metastasis Claim	31
5.1 Unweighted Wilcoxon tests — exploratory	31
5.2 Linear mixed-effects models — three representations	35
Model results tables	38
Three-model forest comparison	40
Model agreement heatmap	42
Claim cluster marginal means	44
6. Final Conclusions	47
6.1 TC cluster reliability verdict	47
6.2 Verdict on the BM metastasis claim	48
6.3 Session info	49

IMPORTANT

Using the new 2026 re-analysis Rds object [here](#).

Overview

This document presents a complete statistical analysis of tumor cell (TC) cluster reliability and tissue-specific enrichment in the MapMet IMC dataset. The dataset comprises imaging mass cytometry (IMC) acquisitions from primary tumors (PT) and bone marrow (BM) samples across multiple timepoints from neuroblastoma patients.

The analysis addresses two central questions:

1. **Which TC cluster calls are statistically reliable**, given that some samples — particularly BM samples — contain very few TC cells? TC cluster proportions derived from small cell counts are subject to substantial multinomial sampling noise and may not reflect true biology.
2. **Are specific TC clusters genuinely enriched in BM relative to PT at diagnosis**, as claimed for marker-lo TC and early SYM-like TC cells?

Study design note: The cohort has a mixed paired/unpaired structure. A subset of patients contributed both a primary tumor (PT) and a bone marrow (BM) biopsy at the same timepoint — these paired samples enable within-patient BM vs PT comparisons that control for inter-patient variability. The remainder contributed only one tissue type. The statistical models in section 5 account for this structure explicitly via a random intercept per patient.

The analysis proceeds in four stages: (1) bootstrap-based stability assessment of cluster calls, (2) a three-test artifact detection framework, (3) validation of flagged clusters in PT samples, and (4) formal statistical testing of BM vs PT enrichment using multiple complementary models.

1. Setup and Data Loading

1.1 Libraries and parameters

All key thresholds are defined here and propagate through the entire analysis. Adjusting these values here automatically updates all downstream results.

```
library(tidyverse)
library(ggrepel)
library(patchwork)
library(lme4)
library(lmerTest)
library(emmeans)

set.seed(42)

metadata <- read.table(
  "/home/rstudio/mnt_data/MapMet/lazic.imc.2024/metadata/20231128_metadata_complete.csv",
  sep = ",", header = TRUE, row.names = 1
)
imc <- readRDS("/home/rstudio/mnt_out/MapMet/Rds/imc_v2026.rds")
imc_metadata <- read.table(
  "/home/rstudio/mnt_out/MapMet/tables/imc_metadata.csv",
  sep = ",", header = TRUE, row.names = 1)

library(Seurat)
imc <- AddMetaData(imc, imc_metadata)

seu_meta <- imc@meta.data

# Verify the key columns are present
required_cols <- c("sample_id", "tissue", "celltype")
missing <- setdiff(required_cols, colnames(seu_meta))
if (length(missing) > 0) {
  message("Missing columns: ", paste(missing, collapse = ", "))
} else {
  message("All columns present")
}

head(seu_meta[, required_cols])
```

```
##                                     sample_id tissue
## 20220519_20211222_03-014_BM_005_1440 20220519_20211222_03-014_BM_005    BM
## 20220519_20211222_03-014_BM_005_1443 20220519_20211222_03-014_BM_005    BM
## 20220519_20211222_03-014_BM_005_1453 20220519_20211222_03-014_BM_005    BM
## 20220519_20211222_03-014_BM_005_1455 20220519_20211222_03-014_BM_005    BM
## 20220519_20211222_03-014_BM_005_1456 20220519_20211222_03-014_BM_005    BM
## 20220519_20211222_03-014_BM_005_1457 20220519_20211222_03-014_BM_005    BM
##                                     celltype
## 20220519_20211222_03-014_BM_005_1440 CHGAhi TC
## 20220519_20211222_03-014_BM_005_1443 CHGAhi TC
## 20220519_20211222_03-014_BM_005_1453 CHGAhi TC
## 20220519_20211222_03-014_BM_005_1455 CHGAhi TC
```

```
## 20220519_20211222_03-014_BM_005_1456 PRPHhi TC
## 20220519_20211222_03-014_BM_005_1457 CHGAhi TC
```

```
TC_PATTERN      <- "TC|[Tt]umor" # regex to identify TC celltypes
BM_LABEL        <- "BM"          # tissue label for bone marrow
PT_LABEL        <- "PT"          # tissue label for primary tumor
DE_LABEL        <- "DE"          # timepoint label for diagnosis

N_BOOT          <- 1000          # bootstrap resamples per unit
MIN_TC_CELLS    <- 30           # minimum pooled TC cells for reliability flag
MAX_MEAN_SD     <- 0.10         # mean bootstrap SD threshold for instability flag

MIN_DETECT_CELLS <- 3           # minimum cells for cluster detection (T2/T3)
MIN_DETECT_PROP  <- 0.01        # minimum proportion for detection (1%)
MIN_DETECT_CELLS_T2 <- 1       # relaxed threshold for T2 presence test
CLR_PSEUDOCOUNT <- 0.5          # Haldane-Anscombe correction before log transform

# Colour palette for reliability tiers
rel_colours <- c(
  "critical (n<10)"   = "#E24B4A",
  "unreliable (n<30)" = "#EF9F27",
  "marginal (n<100)"  = "#BA7517",
  "reliable (n 100)"  = "#185FA5"
)
```

1.2 Metadata preparation

Each cell in `seu_meta` carries a `sample_id` encoding the patient ID, tissue type, and acquisition number. The patient-level ID is extracted via regex and joined to the external metadata table to attach `tissue` and `Timepoint` annotations.

The join is performed on both `Sample` (patient ID) and `tissue` (tissue type) simultaneously to prevent the many-to-many row explosion that occurs when a patient has samples from multiple tissues — each cell is matched only to its own tissue’s metadata entry.

A subset of BM samples come from stage L patients (no tumor infiltration detected) who were enrolled as controls at the DE timepoint. These samples have no explicit `Timepoint` entry in the metadata table; they are assigned `DE_LABEL` here before any downstream filtering.

```
# name of TC clusters
grep("TC|[Tt]umor", unique(seu_meta$celltype), value = TRUE)
```

```
## [1] "CHGAhi TC"           "PRPHhi TC"
## [3] "early NB TC"         "marker-lo TC"
## [5] "CXCR4hi TC"          "CD56+ TC"
## [7] "Ki67hi TC"           "GD2hi TC"
## [9] "ELAVL4hi GATA3hi SOX10hi TC" "LUMhi TC"
## [11] "SOX10hi ELAVL4hi GATA3hi TC"
```

```
#
unique(seu_meta$tissue)
```

```
## [1] "BM" "PT" "AD"
```

```
unique(seu_meta$timepoint)
```

```
## [1] "DE"
```

```
unique(seu_meta$control)
```

```
## [1] "no" NA
```

```
seu_meta <- seu_meta %>%  
  filter(tissue %in% c(BM_LABEL, PT_LABEL))  
  
seu_meta <- seu_meta %>%  
  rename(  
    Sample = patient_id,  
    tissue = tissue,  
    Timepoint = timepoint  
  ) %>%  
  mutate(  
    is_control = is.na(control),  
    Timepoint = ifelse(is_control & is.na(Timepoint), DE_LABEL, Timepoint)  
  )
```

1.3 DE cohort overview

The figure below shows the structure of the DE timepoint cohort — the cohort used for all BM vs PT enrichment analyses. Each row is one patient. Coloured tiles indicate which tissue types are available for that patient. Paired patients (with both a BM and a PT sample at DE) are shown first; stage L BM controls are indicated separately.

```
de_cohort <- seu_meta %>%  
  filter(Timepoint == DE_LABEL) %>%  
  distinct(Sample, tissue, is_control)  
  
paired_patients <- de_cohort %>%  
  count(Sample) %>%  
  filter(n == 2) %>%  
  pull(Sample)  
  
bm_only_patients <- de_cohort %>%  
  filter(tissue == BM_LABEL, !Sample %in% paired_patients) %>%  
  pull(Sample) %>% unique()  
  
pt_only_patients <- de_cohort %>%  
  filter(tissue == PT_LABEL, !Sample %in% paired_patients) %>%  
  pull(Sample) %>% unique()  
  
patient_order <- c(  
  sort(paired_patients),
```

```

    sort(pt_only_patients),
    sort(bm_only_patients)
  )

cohort_df <- de_cohort %>%
  mutate(
    Sample      = factor(Sample, levels = rev(patient_order)),
    tile_label  = case_when(
      is_control ~ "BM (stage L control)",
      tissue == BM_LABEL ~ "BM (neuroblastoma)",
      tissue == PT_LABEL ~ "PT"
    ),
    paired_group = case_when(
      Sample %in% paired_patients ~ "Paired",
      Sample %in% pt_only_patients ~ "PT only",
      TRUE ~ "BM only"
    )
  )

tile_colours <- c(
  "BM (neuroblastoma)" = "#E24B4A",
  "BM (stage L control)" = "#F0997B",
  "PT" = "#185FA5"
)

group_counts <- tibble(
  paired_group = c("Paired", "PT only", "BM only"),
  label = c(
    paste0("Paired (n=", length(paired_patients), ")"),
    paste0("PT only (n=", length(pt_only_patients), ")"),
    paste0("BM only (n=", length(bm_only_patients), ")")
  )
)

group_y_positions <- tibble(
  Sample      = factor(patient_order, levels = rev(patient_order)),
  paired_group = case_when(
    patient_order %in% paired_patients ~ "Paired",
    patient_order %in% pt_only_patients ~ "PT only",
    TRUE ~ "BM only"
  )
) %>%
  group_by(paired_group) %>%
  summarise(
    y_mid = median(as.numeric(Sample)),
    .groups = "drop"
  ) %>%
  left_join(group_counts, by = "paired_group")

p_cohort <- ggplot(cohort_df,
  aes(x = tissue, y = Sample, fill = tile_label)) +
  geom_tile(colour = "white", linewidth = 0.8, width = 0.85, height = 0.85) +
  geom_segment(

```

```

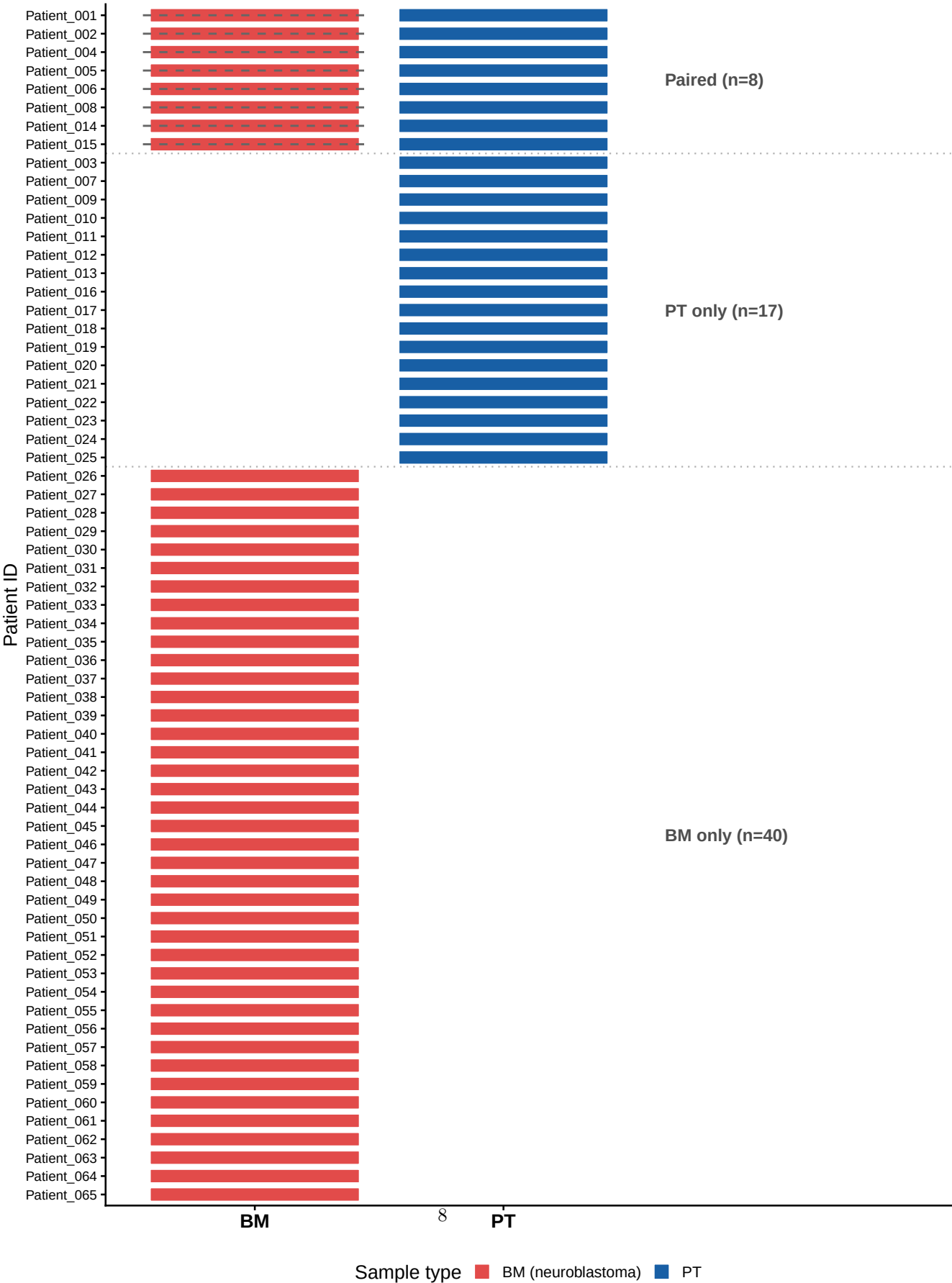
data = cohort_df %>%
  filter(Sample %in% paired_patients) %>%
  distinct(Sample),
aes(x = 0.55, xend = 1.45, y = Sample, yend = Sample),
colour = "grey40", linewidth = 0.5, linetype = "dashed",
inherit.aes = FALSE
) +
geom_text(
  data = group_y_positions,
  aes(x = 2.65, y = y_mid, label = label),
  hjust = 0, size = 3.2, colour = "grey30", fontface = "bold",
  inherit.aes = FALSE
) +
geom_hline(
  yintercept = c(
    length(bm_only_patients) + 0.5,
    length(bm_only_patients) + length(pt_only_patients) + 0.5
  ),
  colour = "grey70", linewidth = 0.4, linetype = "dotted"
) +
scale_fill_manual(values = tile_colours, name = "Sample type") +
scale_x_discrete(expand = expansion(add = c(0.6, 1.8))) +
labs(
  title = "DE cohort - sample availability per patient",
  subtitle = paste0(
    "Each row = one patient | dashed lines connect paired BM+PT samples\n",
    "Total: ", length(patient_order), " patients | ",
    length(paired_patients), " paired | ",
    length(pt_only_patients), " PT only | ",
    length(bm_only_patients), " BM only (",
    sum(de_cohort$is_control[!duplicated(de_cohort$Sample)]), " stage L controls)"
  ),
  x = NULL,
  y = "Patient ID"
) +
theme_classic(base_size = 10) +
theme(
  plot.title = element_text(size = 12, face = "bold", colour = "black"),
  plot.subtitle = element_text(size = 8.5, colour = "grey35"),
  axis.text.x = element_text(size = 10, colour = "black", face = "bold"),
  axis.text.y = element_text(size = 7, colour = "black"),
  legend.position = "bottom",
  legend.key.size = unit(0.4, "cm")
)

```

p_cohort

DE cohort – sample availability per patient

Each row = one patient | dashed lines connect paired BM+PT samples
Total: 65 patients | 8 paired | 17 PT only | 40 BM only (0 stage L controls)



2.1 Rationale

IMC data from BM samples at diagnosis frequently contains very few tumor cells — in some acquisitions, fewer than 10 TC cells are identified across all ROIs. When TC cluster proportions are derived from such small counts, the resulting composition reflects multinomial sampling noise rather than true biology.

To quantify this uncertainty, I use **non-parametric bootstrap resampling**: for each patient $\times$ tissue $\times$ timepoint unit, TC cells are resampled with replacement 1,000 times, and cluster proportions are recomputed each time. The spread of these resampled proportions — measured as the bootstrap standard deviation and 95% confidence interval width — directly quantifies how reliably any given cluster proportion can be estimated from the available cell count.

The key insight is the theoretical $1/\sqrt{n}$ relationship between cell count and proportion uncertainty: with 10 TC cells, the 95% CI on a cluster at 20% spans roughly 0–55%; with 100 cells it narrows to 12–32%; with 500 cells it reaches 16–24%. Samples below ~30 TC cells are therefore treated as unreliable for composition analysis.

2.2 Aggregation unit definition

The analysis unit is **patient $\times$ tissue $\times$ timepoint** — all ROIs from the same patient in the same tissue at the same timepoint are pooled before bootstrapping. This reflects the true biological unit of interest and avoids pseudo-replication at the ROI level.

```
# Identify TC clusters
tc_cols <- grep(TC_PATTERN, unique(seu_meta$celltype), value = TRUE)
message("TC clusters: ", paste(tc_cols, collapse = ", "))

CLUSTERS_OF_INTEREST <- tc_cols
ARTIFACT_CLUSTERS <- tc_cols

# Build TC-only cell table with unit IDs
tc_meta_agg <- seu_meta %>%
  filter(grepl(TC_PATTERN, celltype)) %>%
  select(Sample, tissue, Timepoint, sample_id, TC_cluster = celltype) %>%
  mutate(unit_id = paste0(Sample, " | ", tissue, " | ", Timepoint))

# ROIs pooled per unit
rois_per_unit <- seu_meta %>%
  mutate(unit_id = paste0(Sample, " | ", tissue, " | ", Timepoint)) %>%
  distinct(unit_id, sample_id) %>%
  count(unit_id, name = "n_ROIs")

# Per-unit totals and per-cluster counts
tc_counts_agg <- tc_meta_agg %>% count(unit_id, name = "n_TC_cells")
tc_counts_per_cluster <- tc_meta_agg %>% count(unit_id, TC_cluster, name = "n_cluster_cells")
observed_props_agg <- tc_meta_agg %>%
  count(unit_id, TC_cluster) %>%
  group_by(unit_id) %>%
  mutate(observed_prop = n / sum(n)) %>%
  ungroup()

tc_meta_agg %>%
  distinct(unit_id, Sample, tissue, Timepoint) %>%
```

```
count(tissue, Timepoint) %>%
knitr::kable(caption = "Number of units per tissue × timepoint")
```

Table 1: Number of units per tissue × timepoint

tissue	Timepoint	n
BM	DE	48
PT	DE	25

2.3 Bootstrap resampling

```
run_bootstrap_unit <- function(uid, tc_data, n_boot, all_clusters) {
  cells <- tc_data %>% filter(unit_id == uid) %>% pull(TC_cluster)
  n <- length(cells)
  resamples <- replicate(n_boot, {
    s <- sample(cells, size = n, replace = TRUE)
    props <- table(factor(s, levels = all_clusters)) / n
    as.numeric(props)
  })
  tibble(
    unit_id = uid,
    TC_cluster = all_clusters,
    boot_mean = rowMeans(resamples),
    boot_sd = apply(resamples, 1, sd),
    boot_q025 = apply(resamples, 1, quantile, 0.025),
    boot_q975 = apply(resamples, 1, quantile, 0.975),
    boot_q25 = apply(resamples, 1, quantile, 0.25),
    boot_q75 = apply(resamples, 1, quantile, 0.75),
    n_TC_cells = n
  )
}

message("Bootstrapping ", n_distinct(tc_meta_agg$unit_id),
       " units × ", N_BOOT, " iterations...")

boot_results_agg <- map_dfr(
  unique(tc_meta_agg$unit_id),
  run_bootstrap_unit,
  tc_data = tc_meta_agg,
  n_boot = N_BOOT,
  all_clusters = tc_cols
)
```

2.4 Instability scores and flagging

Each unit receives two flags: (1) `flag_n` if fewer than 30 TC cells are pooled, and (2) `flag_sd` if the mean bootstrap SD across clusters exceeds 0.10. A unit is flagged if either criterion fires.

```

instability_agg <- boot_results_agg %>%
  group_by(unit_id, n_TC_cells) %>%
  summarise(
    mean_boot_SD = mean(boot_sd),
    mean_CI_width = mean(boot_q975 - boot_q025),
    max_CI_width = max(boot_q975 - boot_q025),
    .groups = "drop"
  ) %>%
  separate(unit_id, into = c("Sample", "tissue", "Timepoint"),
    sep = " \\| ", remove = FALSE) %>%
  left_join(rois_per_unit, by = "unit_id") %>%
  mutate(
    flag_n = n_TC_cells < MIN_TC_CELLS,
    flag_sd = mean_boot_SD > MAX_MEAN_SD,
    flagged = flag_n | flag_sd,
    flag_reason = case_when(
      flag_n & flag_sd ~ paste0("n=", n_TC_cells, " & SD>", MAX_MEAN_SD),
      flag_n ~ paste0("n=", n_TC_cells, " < ", MIN_TC_CELLS),
      flag_sd ~ paste0("mean SD=", round(mean_boot_SD, 3)),
      TRUE ~ "OK"
    ),
    reliability = case_when(
      n_TC_cells < 10 ~ "critical (n<10)",
      n_TC_cells < 30 ~ "unreliable (n<30)",
      n_TC_cells < 100 ~ "marginal (n<100)",
      TRUE ~ "reliable (n 100)"
    ) %>% factor(levels = c("critical (n<10)", "unreliable (n<30)",
      "marginal (n<100)", "reliable (n 100)")),
    is_BM = tissue == BM_LABEL,
    is_DE = Timepoint == DE_LABEL,
    is_BM_DE = is_BM & is_DE,
    group = case_when(
      is_BM_DE ~ "BM + DE",
      is_BM & !is_DE ~ "BM, other timepoint",
      !is_BM & is_DE ~ "Tumor + DE",
      TRUE ~ "Tumor, other timepoint"
    ) %>% factor(levels = c("BM + DE", "BM, other timepoint",
      "Tumor + DE", "Tumor, other timepoint"))
  )

instability_agg %>%
  count(group, reliability) %>%
  knitr::kable(caption = "Unit counts by group and reliability tier")

```

Table 2: Unit counts by group and reliability tier

group	reliability	n
BM + DE	unreliable (n<30)	1
BM + DE	marginal (n<100)	1
BM + DE	reliable (n 100)	46
Tumor + DE	reliable (n 100)	25

2.5 Instability plot — BM+DE units highlighted

The plot below shows the key diagnostic: mean bootstrap SD (y-axis) against pooled TC cell count (x-axis, log scale) for all units. The expected $1/\sqrt{n}$ decay is visible. BM+DE units are highlighted with bold outlines; dot size reflects the number of ROIs pooled.

All units

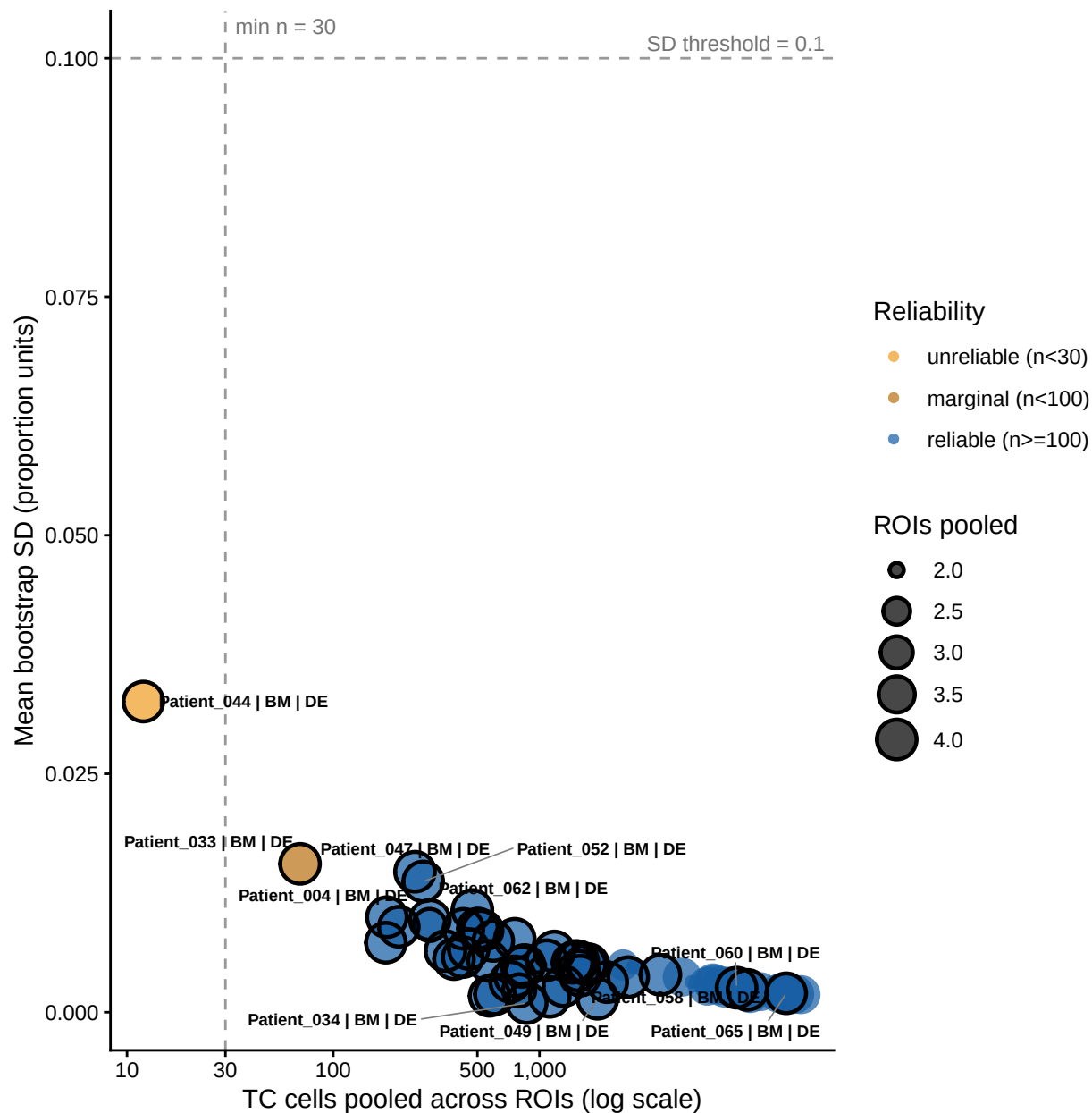
```
bm_de_units <- instability_agg %>% filter(is_BM_DE)

p_highlighted <- ggplot(instability_agg,
  aes(x = n_TC_cells, y = mean_boot_SD, colour = reliability)) +
  geom_vline(xintercept = MIN_TC_CELLS, linetype = "dashed",
    colour = "grey60", linewidth = 0.5) +
  geom_hline(yintercept = MAX_MEAN_SD, linetype = "dashed",
    colour = "grey60", linewidth = 0.5) +
  geom_point(aes(size = n_ROIs), alpha = 0.72) +
  geom_point(data = bm_de_units, aes(size = n_ROIs),
    shape = 21, stroke = 1.2, colour = "black", fill = NA) +
  geom_text_repel(data = bm_de_units, aes(label = unit_id),
    size = 2.5, fontface = "bold", max.overlaps = 30,
    box.padding = 0.4, segment.size = 0.3,
    segment.colour = "grey50", colour = "black") +
  annotate("text", x = MIN_TC_CELLS, y = Inf,
    label = paste0("min n = ", MIN_TC_CELLS),
    hjust = -0.1, vjust = 1.5, size = 3, colour = "grey45") +
  annotate("text", x = Inf, y = MAX_MEAN_SD,
    label = paste0("SD threshold = ", MAX_MEAN_SD),
    hjust = 1.05, vjust = -0.5, size = 3, colour = "grey45") +
  scale_colour_manual(values = rel_colours, name = "Reliability") +
  scale_size_continuous(range = c(2, 7), name = "ROIs pooled") +
  scale_x_continuous(trans = "log10", breaks = c(1, 5, 10, 30, 100, 500, 1000),
    labels = scales::comma) +
  labs(
    title = "TC cluster call stability - BM+DE units highlighted",
    subtitle = paste0(sum(bm_de_units$flagged), " of ", nrow(bm_de_units),
      " BM+DE units flagged | one dot = patient × tissue × timepoint | dot size = ROIs pooled"),
    x = "TC cells pooled across ROIs (log scale)",
    y = "Mean bootstrap SD (proportion units)")
  ) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
    plot.subtitle = element_text(size = 9, colour = "grey35"),
    axis.text = element_text(colour = "black"),
    legend.position = "right")

p_highlighted
```

TC cluster call stability – BM+DE units highlighted

1 of 48 BM+DE units flagged | one dot = patient × tissue × timepoint | dot size = ROIs pooled



BM+DE bootstrap ribbons

Bootstrap confidence intervals per TC cluster for BM+DE units, sorted by ascending TC cell count. Flagged units are marked with . Wide ribbons with high uncertainty should not be interpreted biologically.

```
bm_de_unit_ids <- bm_de_units %>% arrange(n_TC_cells) %>% pull(unit_id)

plot_df_bm_de <- boot_results_agg %>%
  filter(unit_id %in% bm_de_unit_ids) %>%
  select(-n_TC_cells) %>%
```

```

left_join(observed_props_agg %>% select(unit_id, TC_cluster, observed_prop),
          by = c("unit_id", "TC_cluster")) %>%
left_join(instability_agg %>% select(unit_id, flagged, reliability, n_ROIs, n_TC_cells),
          by = "unit_id") %>%
replace_na(list(observed_prop = 0)) %>%
mutate(
  unit_label = paste0(ifelse(flagged, " ", ""), unit_id,
                        "\n(", n_ROIs, " ROIs, n=", n_TC_cells, ")"),
  unit_label = factor(unit_label,
                      levels = unique(unit_label[order(match(unit_id, bm_de_unit_ids))]))
)

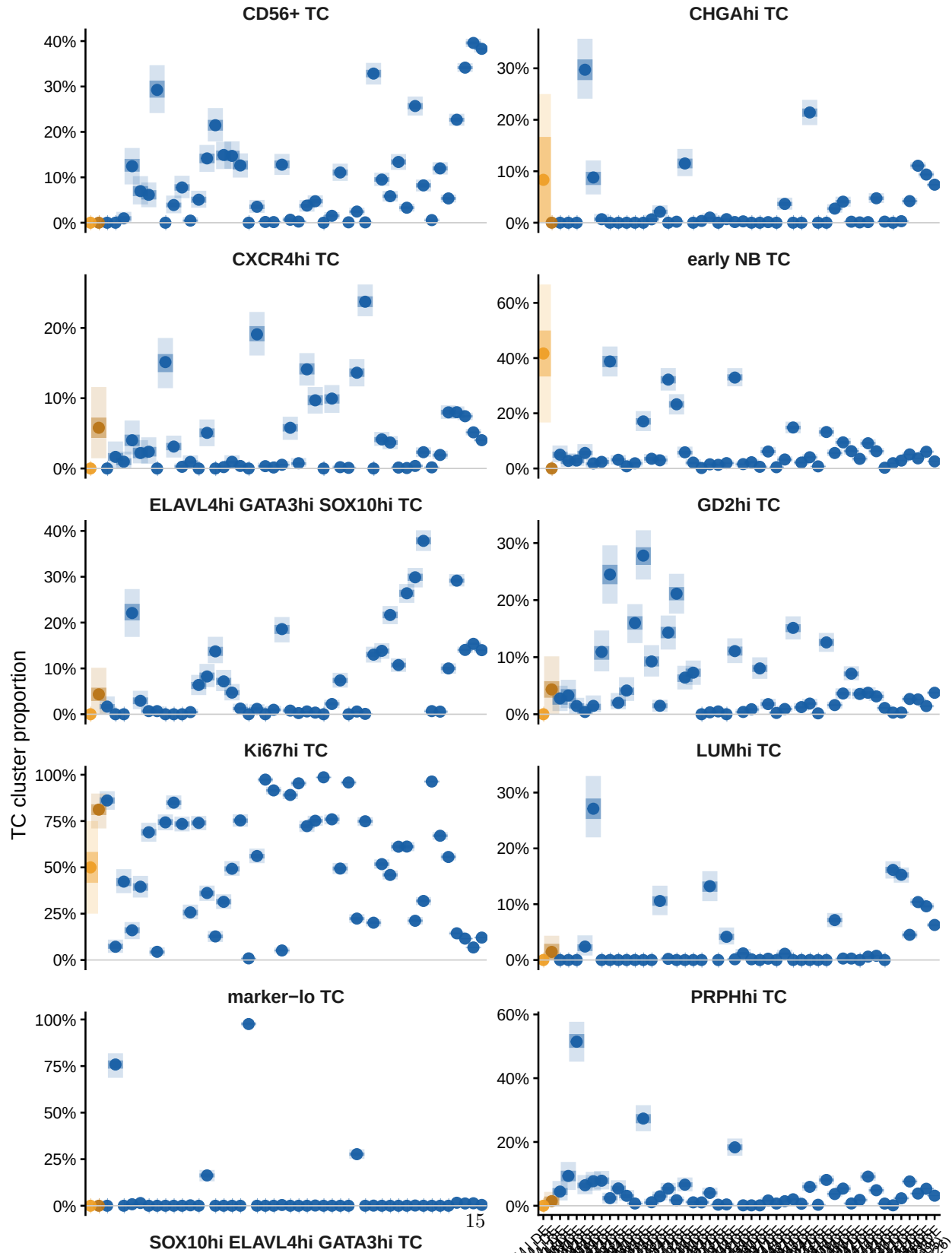
ggplot(plot_df_bm_de, aes(x = unit_label, colour = reliability, fill = reliability)) +
  geom_linerange(aes(ymin = boot_q025, ymax = boot_q975), linewidth = 3.5, alpha = 0.18) +
  geom_linerange(aes(ymin = boot_q25, ymax = boot_q75), linewidth = 3.5, alpha = 0.45) +
  geom_point(aes(y = observed_prop), size = 2.2, alpha = 0.95) +
  geom_hline(yintercept = 0, colour = "grey80", linewidth = 0.3) +
  facet_wrap(~ TC_cluster, scales = "free_y", ncol = 2) +
  scale_colour_manual(values = rel_colours, name = "Reliability") +
  scale_fill_manual(values = rel_colours, name = "Reliability") +
  scale_y_continuous(labels = scales::percent_format(accuracy = 1)) +
  labs(title = paste0("Bootstrap stability - BM units at ", DE_LABEL, " timepoint"),
       subtitle = "Point = observed; thick = IQR; thin = 95% CI | = flagged | sorted low → high pooled",
       x = NULL, y = "TC cluster proportion") +
  theme_classic(base_size = 10) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8, colour = "grey35"),
        axis.text.x = element_text(angle = 45, hjust = 1, size = 7, colour = "black"),
        strip.text = element_text(face = "bold", size = 9),
        strip.background = element_blank(),
        legend.position = "top")

```

Bootstrap stability – BM units at DE timepoint

Point = observed; thick = IQR; thin = 95% CI | . = flagged | sorted low → high pooled TC cells

Reliability ● unreliable (n<30) ● marginal (n<100) ● reliable (n>=100)



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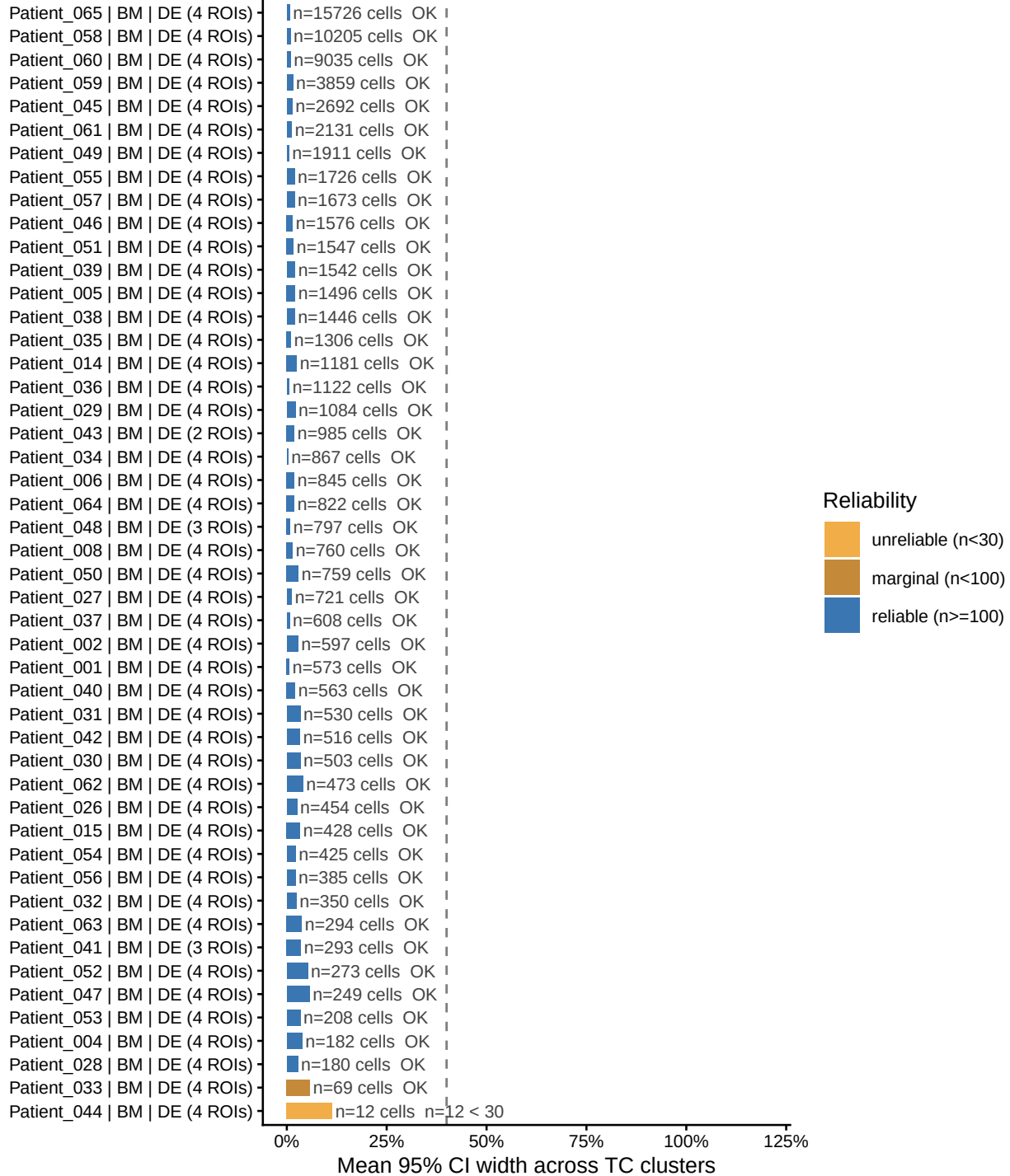
BM+DE CI width summary

```
p_table_order <- instability_agg %>%
  filter(is_BM_DE) %>%
  arrange(n_TC_cells) %>%
  mutate(unit_label = paste0(unit_id, " (", n_ROIs, " ROIs)")) %>%
  pull(unit_label)

instability_agg %>%
  filter(is_BM_DE) %>%
  arrange(n_TC_cells) %>%
  mutate(
    unit_label = paste0(unit_id, " (", n_ROIs, " ROIs)"),
    unit_label = factor(unit_label, levels = p_table_order)
  ) %>%
  ggplot(aes(y = unit_label)) +
  geom_col(aes(x = mean_CI_width, fill = reliability), width = 0.7, alpha = 0.85) +
  geom_vline(xintercept = 0.40, linetype = "dashed", colour = "grey50", linewidth = 0.5) +
  geom_text(aes(x = mean_CI_width + 0.008,
    label = paste0("n=", n_TC_cells, " cells ", flag_reason)),
    hjust = 0, size = 2.8, colour = "grey25") +
  scale_fill_manual(values = rel_colours, name = "Reliability") +
  scale_x_continuous(limits = c(0, 1.2), labels = scales::percent_format(accuracy = 1)) +
  labs(title = "BM + DE units: mean bootstrap 95% CI width",
    subtitle = "Dashed = 40 pp threshold (CI spans nearly half the proportion range - uninterpretable)",
    x = "Mean 95% CI width across TC clusters", y = NULL) +
  theme_classic(base_size = 10) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
    plot.subtitle = element_text(size = 8.5, colour = "grey35"),
    axis.text = element_text(colour = "black"))
```


BM + DE units: mean bootstrap 95% CI width

Dashed = 40 pp threshold (CI spans nearly half the proportion range – uninterpretable)



3. Artifact Detection — Three-Test Framework

The three tests below ask complementary questions about whether a TC cluster's apparent presence in low-TC-count samples reflects biology or sampling noise. A cluster flagged by 2/3 tests is classified as a probable sampling artifact and excluded from BM composition analyses.

3.1 Test 1 — Per-cluster bootstrap instability

Question: Which clusters have the highest bootstrap SD overall, and is that instability concentrated in low-n units (artifact signature) or distributed across reliable units (biological variability signature)?

```
cluster_rank <- boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(instability_agg %>% select(unit_id, n_TC_cells), by = "unit_id") %>%
  group_by(TC_cluster) %>%
  summarise(overall_mean_SD = mean(boot_sd), overall_median_SD = median(boot_sd),
    .groups = "drop") %>%
  arrange(desc(overall_mean_SD)) %>%
  mutate(instability_rank = row_number())

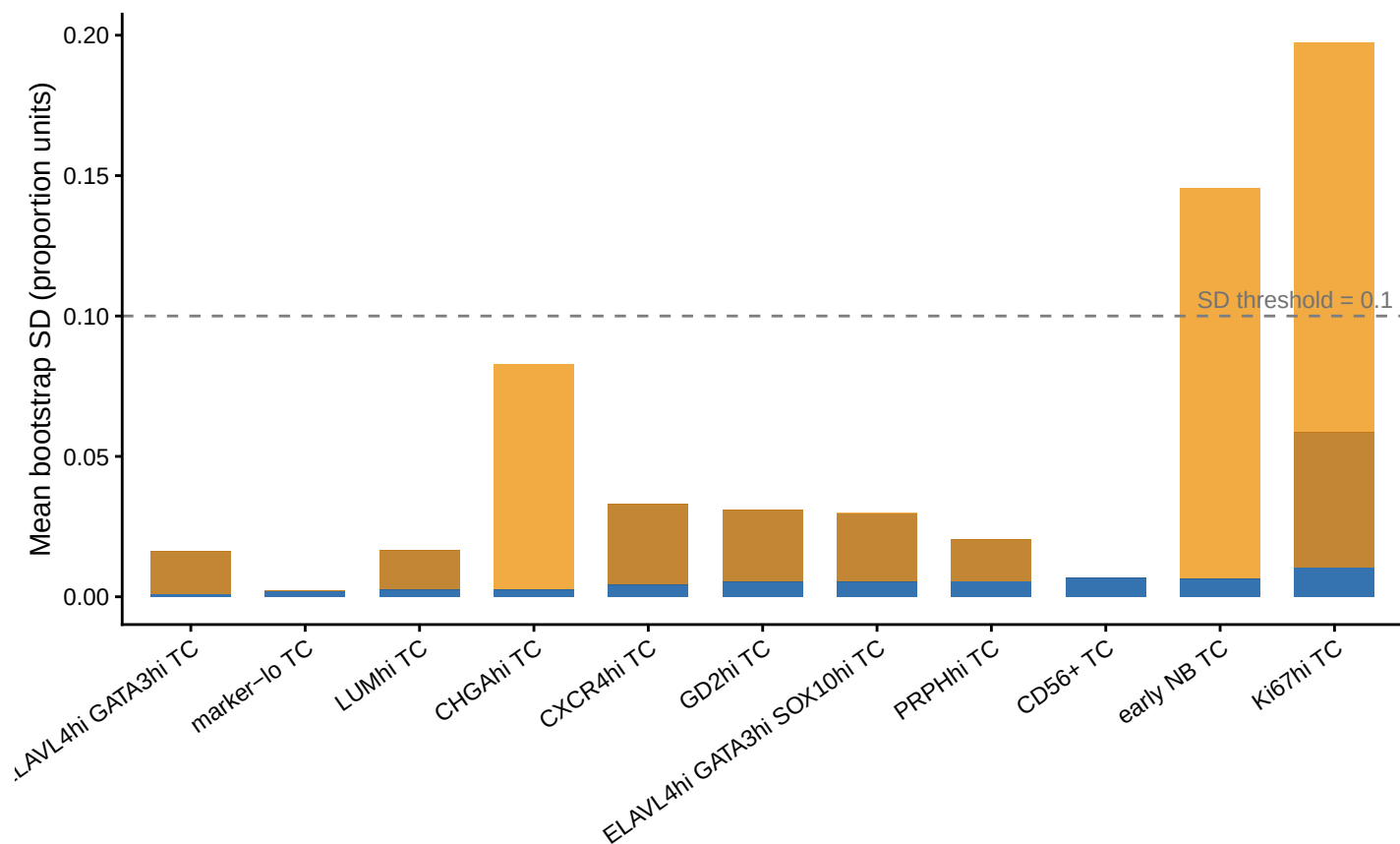
cluster_instability <- boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(instability_agg %>% select(unit_id, reliability, flagged, n_TC_cells),
    by = "unit_id") %>%
  group_by(TC_cluster, reliability) %>%
  summarise(mean_SD = mean(boot_sd), n_units = n(), .groups = "drop")

t1_cluster_order <- cluster_rank %>% arrange(overall_mean_SD) %>% pull(TC_cluster)
```

```
cluster_instability %>%
  left_join(cluster_rank %>% select(TC_cluster, instability_rank), by = "TC_cluster") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t1_cluster_order)) %>%
  ggplot(aes(x = TC_cluster, y = mean_SD, fill = reliability)) +
  geom_col(position = "stack", width = 0.7, alpha = 0.88) +
  geom_hline(yintercept = MAX_MEAN_SD, linetype = "dashed",
    colour = "grey50", linewidth = 0.5) +
  annotate("text", x = Inf, y = MAX_MEAN_SD,
    label = paste0("SD threshold = ", MAX_MEAN_SD),
    hjust = 1.05, vjust = -0.5, size = 3, colour = "grey45") +
  scale_fill_manual(values = rel_colours, name = "Reliability tier") +
  labs(title = "Test 1 - Per-cluster bootstrap instability across all units",
    subtitle = "High SD driven only by unreliable/critical units = artifact; high SD in reliable units = biology",
    x = NULL, y = "Mean bootstrap SD (proportion units)") +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
    plot.subtitle = element_text(size = 8.5, colour = "grey35"),
    axis.text.x = element_text(angle = 35, hjust = 1, colour = "black"),
    axis.text.y = element_text(colour = "black"))
```

Test 1 – Per-cluster bootstrap instability across all units

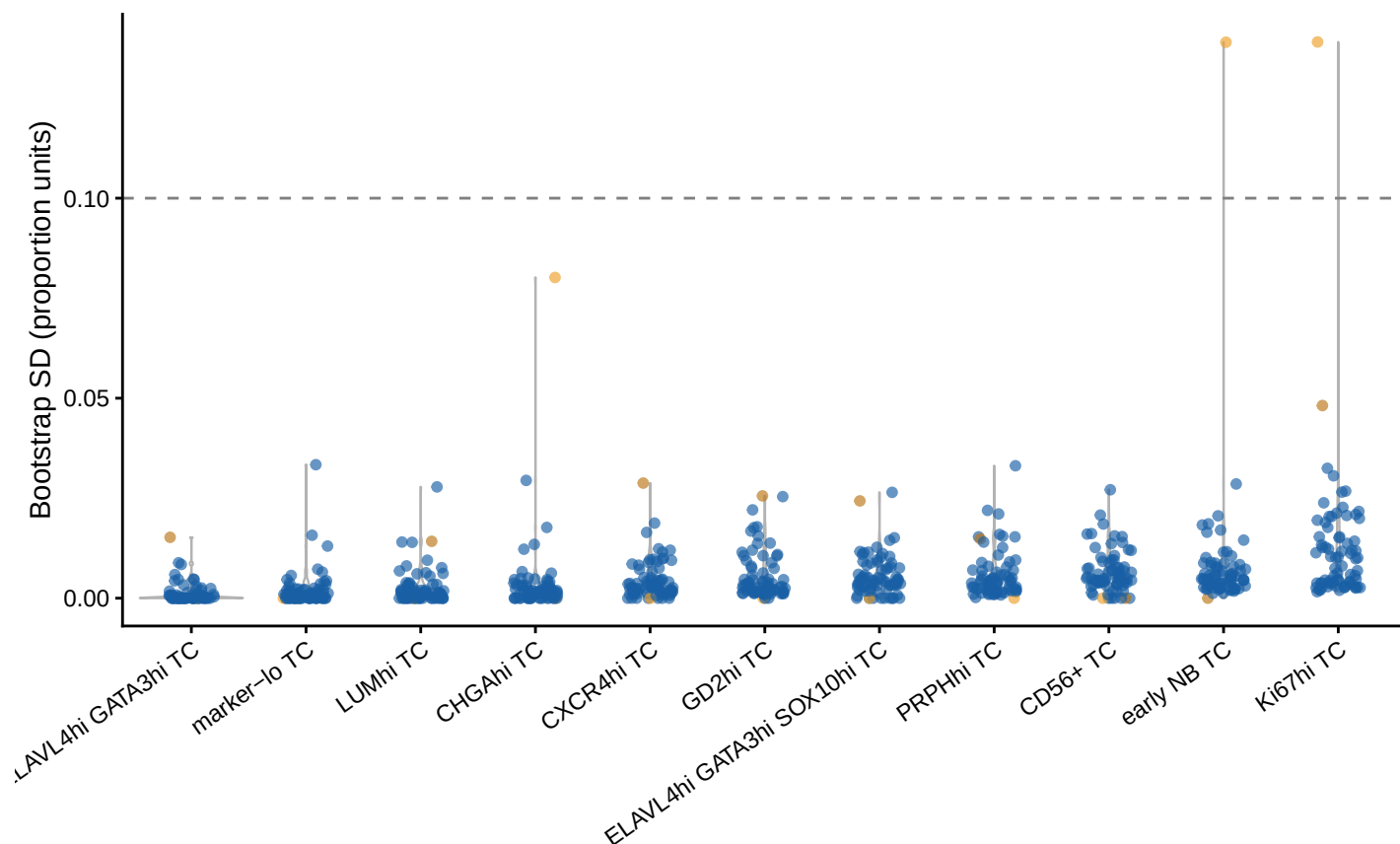
High SD driven only by unreliable/critical units = artifact; high SD in reliable units = variable biology



```
boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(instability_agg %>% select(unit_id, reliability, n_TC_cells), by = "unit_id") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t1_cluster_order)) %>%
  ggplot(aes(x = TC_cluster, y = boot_sd, colour = reliability)) +
  geom_violin(aes(group = TC_cluster), fill = "grey92", colour = "grey70",
    linewidth = 0.4, alpha = 0.6) +
  geom_jitter(width = 0.2, size = 1.4, alpha = 0.65) +
  geom_hline(yintercept = MAX_MEAN_SD, linetype = "dashed",
    colour = "grey50", linewidth = 0.5) +
  scale_colour_manual(values = rel_colours, name = "Reliability") +
  labs(title = "Test 1b - Bootstrap SD distribution per cluster",
    subtitle = "Each dot = one patient × tissue × timepoint unit",
    x = NULL, y = "Bootstrap SD (proportion units)") +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
    plot.subtitle = element_text(size = 8.5, colour = "grey35"),
    axis.text.x = element_text(angle = 35, hjust = 1, colour = "black"),
    axis.text.y = element_text(colour = "black"))
```

Test 1b – Bootstrap SD distribution per cluster

Each dot = one patient × tissue × timepoint unit



3.2 Test 2 — Detection rate analysis

Question: Is a cluster detected more often in low-TC-count (flagged) samples than in high-TC-count (reliable) samples? A cluster that disappears as cell count increases is a phantom cluster — it exists only because with few cells, random assignment places 1–2 stray cells there by chance.

Detection here uses a relaxed threshold (1 cell AND 1% proportion) to preserve sensitivity for the enrichment-in-flagged question. The strict threshold (3 cells AND 1%) is reserved for Test 3.

```
detection_df_t2 <- observed_props_agg %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(n_cluster_cells = 0)) %>%
  left_join(instability_agg %>% select(unit_id, n_TC_cells, flagged, reliability),
            by = "unit_id") %>%
  mutate(
    detected = n_cluster_cells >= MIN_DETECT_CELLS_T2 & observed_prop >= MIN_DETECT_PROP,
    n_bin    = cut(n_TC_cells,
                  breaks = c(0, 10, 30, 100, 500, Inf),
                  labels = c("<10", "10-30", "30-100", "100-500", ">500"),
                  right = TRUE, include.lo1st = TRUE)
  )
```

```

detection_rate <- detection_df_t2 %>%
  group_by(TC_cluster, n_bin, .drop = FALSE) %>%
  summarise(n_units = n(), n_detected = sum(detected),
            detection_rate = n_detected / n_units, .groups = "drop")

artifact_cor <- detection_df_t2 %>%
  group_by(TC_cluster) %>%
  summarise(
    spearman_r = cor(n_TC_cells, as.numeric(detected),
                     method = "spearman", use = "complete.obs"),
    p_value = tryCatch(
      cor.test(n_TC_cells, as.numeric(detected), method = "spearman",
               exact = FALSE)$p.value, error = function(e) NA_real_),
    pct_detected_flagged = mean(detected[flagged == TRUE], na.rm = TRUE) * 100,
    pct_detected_reliable = mean(detected[flagged == FALSE], na.rm = TRUE) * 100,
    enrichment_in_flagged = pct_detected_flagged - pct_detected_reliable,
    median_cells_when_detected = median(n_cluster_cells[detected], na.rm = TRUE),
    .groups = "drop"
  ) %>%
  mutate(
    p_adj = p.adjust(p_value, method = "BH"),
    artifact_label = case_when(
      spearman_r < -0.3 & p_adj < 0.05 ~ "likely artifact",
      spearman_r < 0 & p_adj < 0.05 ~ "possible artifact",
      spearman_r < 0 & p_adj >= 0.05 ~ "inconclusive",
      TRUE ~ "likely biology"
    ) %>% factor(levels = c("likely artifact", "possible artifact",
                           "inconclusive", "likely biology"))
  ) %>%
  arrange(spearman_r)

t2_heat_order <- artifact_cor %>% arrange(spearman_r) %>% pull(TC_cluster)
t2_enrich_order <- artifact_cor %>% arrange(enrichment_in_flagged) %>% pull(TC_cluster)

artifact_colours <- c(
  "likely artifact" = "#E24B4A", "possible artifact" = "#EF9F27",
  "inconclusive" = "#888780", "likely biology" = "#185FA5"
)

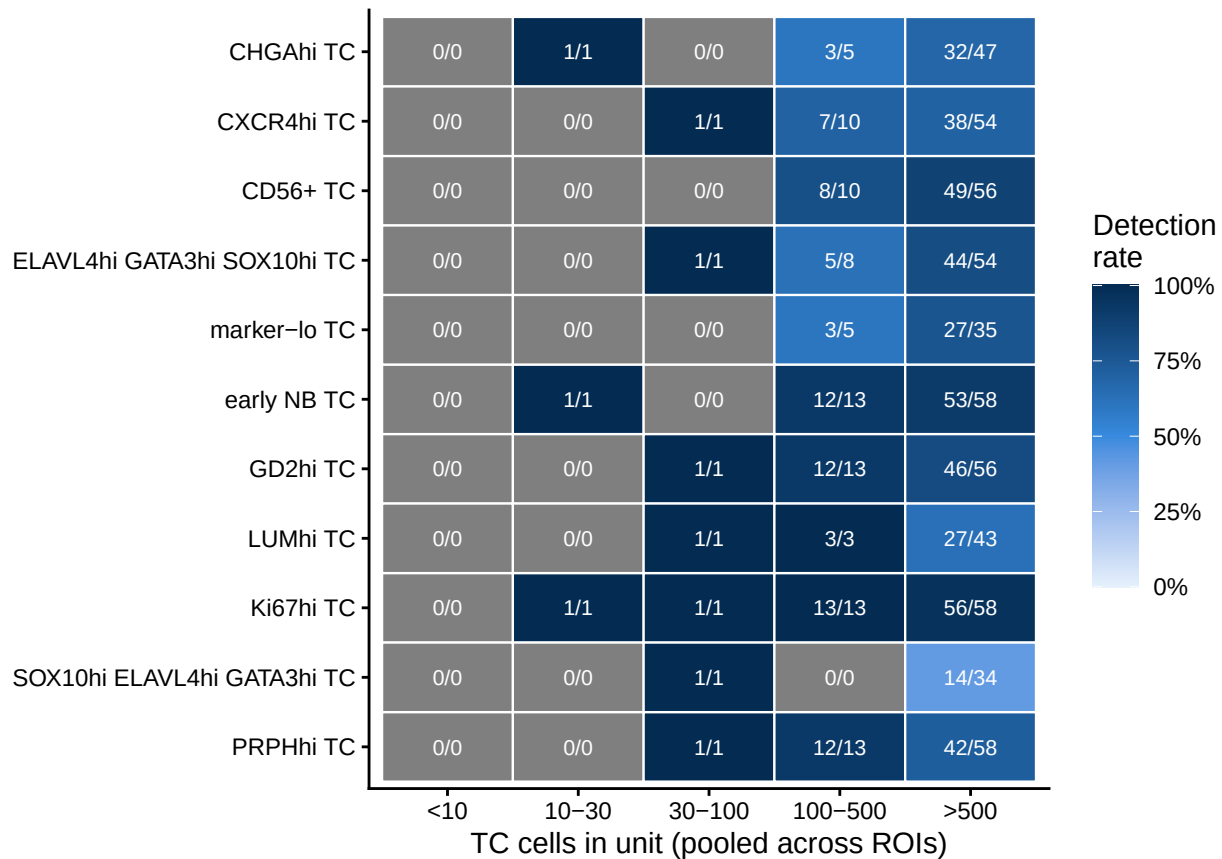
detection_rate %>%
  left_join(artifact_cor %>% select(TC_cluster, artifact_label, spearman_r),
            by = "TC_cluster") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t2_heat_order)) %>%
  ggplot(aes(x = n_bin, y = TC_cluster, fill = detection_rate)) +
  geom_tile(colour = "white", linewidth = 0.4) +
  geom_text(aes(label = paste0(n_detected, "/", n_units)), size = 2.8, colour = "white") +
  scale_fill_gradientn(colours = c("#E6F1FB", "#378ADD", "#042C53"),
                      limits = c(0, 1), labels = scales::percent_format(accuracy = 1),
                      name = "Detection\nrate") +
  labs(title = "Test 2 - Cluster detection rate by n_TC_cells bin",
       subtitle = paste0("Detected = ", MIN_DETECT_CELLS_T2, " cell AND ",
                        MIN_DETECT_PROP * 100, "% | top rows = artifact-like"),
       x = "TC cells in unit (pooled across ROIs)", y = NULL) +

```

```
theme_classic(base_size = 11) +
theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
      plot.subtitle = element_text(size = 8.5, colour = "grey35"),
      axis.text = element_text(colour = "black"),
      legend.key.height = unit(0.8, "cm"))
```

Test 2 – Cluster detection rate by n_TC_cells bin

Detected = >=1 cell AND >=1% | top rows = artifact-like

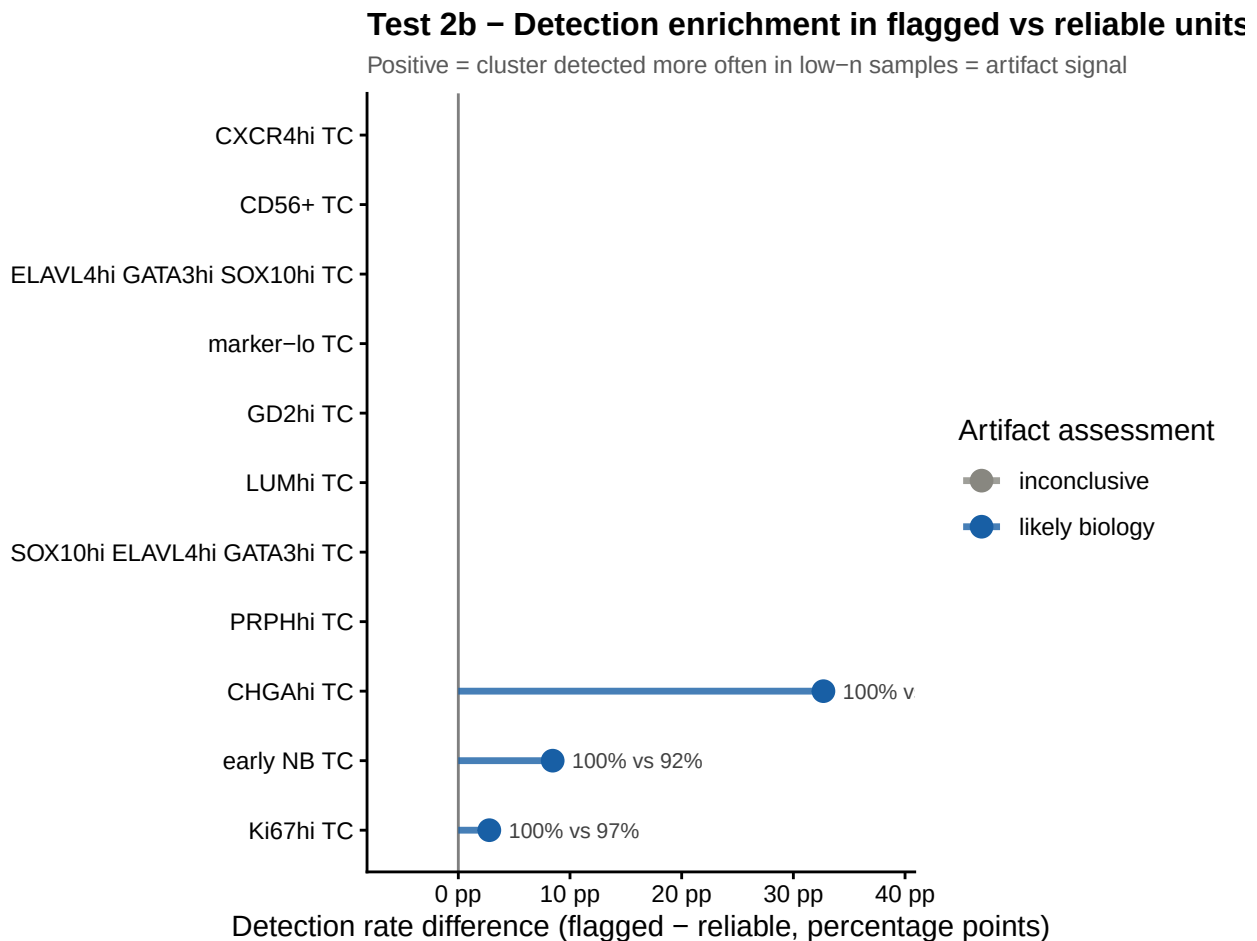


```
artifact_cor %>%
mutate(
  TC_cluster = factor(TC_cluster, levels = t2_enrich_order),
  hjust_val = ifelse(enrichment_in_flagged > 0, -0.15, 1.15)
) %>%
ggplot(aes(y = TC_cluster, colour = artifact_label)) +
geom_vline(xintercept = 0, colour = "grey50", linewidth = 0.5) +
geom_segment(aes(x = 0, xend = enrichment_in_flagged, yend = TC_cluster),
             linewidth = 1.2, alpha = 0.8) +
geom_point(aes(x = enrichment_in_flagged), size = 3.5) +
geom_text(aes(x = enrichment_in_flagged,
              label = paste0(round(pct_detected_flagged, 0), "% vs ",
                              round(pct_detected_reliable, 0), "%"),
              hjust = hjust_val), size = 2.8, colour = "grey25") +
scale_colour_manual(values = artifact_colours, name = "Artifact assessment") +
scale_x_continuous(labels = function(x) paste0(x, " pp"),
```

```

expand = expansion(mult = 0.25)) +
labs(title = "Test 2b - Detection enrichment in flagged vs reliable units",
      subtitle = "Positive = cluster detected more often in low-n samples = artifact signal",
      x = "Detection rate difference (flagged - reliable, percentage points)", y = NULL) +
theme_classic(base_size = 11) +
theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
      plot.subtitle = element_text(size = 8.5, colour = "grey35"),
      axis.text = element_text(colour = "black"))

```



3.3 Test 3 — Bootstrap CI overlap with zero

Question: Even when a cluster is detected (using the strict 3 cells AND 1% threshold), is its bootstrap 95% CI statistically distinguishable from zero? A cluster whose CI routinely includes zero in flagged samples cannot be distinguished from complete absence — its observed proportion could easily be a sampling fluke.

```

zero_overlap <- boot_results_agg %>%
  select(-n_TC_cells) %>%
  left_join(observed_props_agg %>% select(unit_id, TC_cluster, observed_prop),
            by = c("unit_id", "TC_cluster")) %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(observed_prop = 0, n_cluster_cells = 0)) %>%

```

```

left_join(instability_agg %>% select(unit_id, n_TC_cells, flagged, reliability),
          by = "unit_id") %>%
mutate(
  CI_overlaps_zero = boot_q025 <= 0,
  detected          = n_cluster_cells >= MIN_DETECT_CELLS & observed_prop >= MIN_DETECT_PROP
)

zero_summary <- zero_overlap %>%
  filter(detected) %>%
  group_by(TC_cluster) %>%
  summarise(
    n_detected          = n(),
    pct_CI_overlaps_zero = mean(CI_overlaps_zero) * 100,
    pct_flagged_overlap  = ifelse(sum(flagged) > 0,
                                  mean(CI_overlaps_zero[flagged]) * 100, NA_real_),
    pct_reliable_overlap = ifelse(sum(!flagged) > 0,
                                  mean(CI_overlaps_zero[!flagged]) * 100, NA_real_),
    .groups = "drop"
  ) %>%
  left_join(artifact_cor %>% select(TC_cluster, artifact_label, spearman_r),
            by = "TC_cluster") %>%
  arrange(desc(pct_CI_overlaps_zero))

t3_cluster_order <- zero_summary %>% arrange(pct_CI_overlaps_zero) %>% pull(TC_cluster)

```

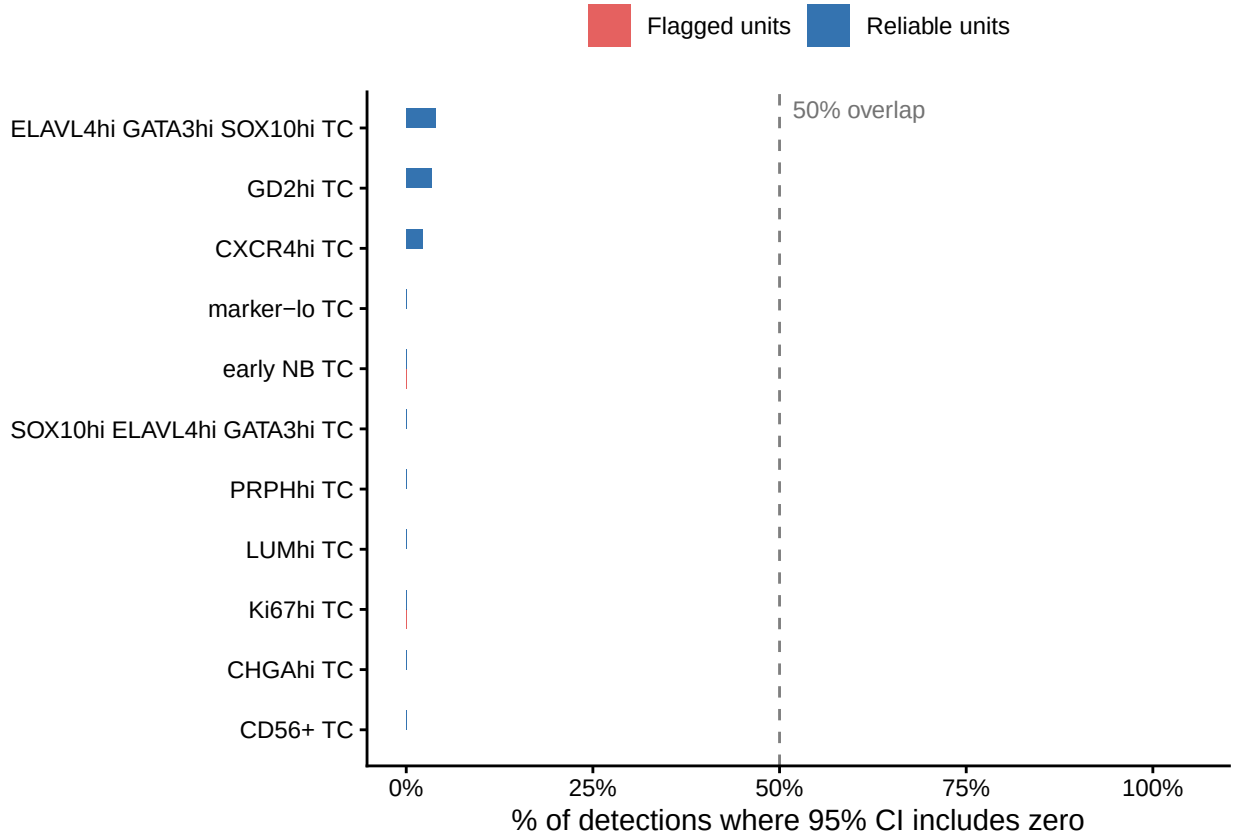
```

zero_summary %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t3_cluster_order)) %>%
  pivot_longer(cols = c(pct_flagged_overlap, pct_reliable_overlap),
               names_to = "unit_type", values_to = "pct_overlap") %>%
  mutate(unit_type = recode(unit_type,
                             pct_flagged_overlap = "Flagged units",
                             pct_reliable_overlap = "Reliable units")) %>%
  ggplot(aes(y = TC_cluster, x = pct_overlap, fill = unit_type)) +
  geom_col(position = "dodge", width = 0.65, alpha = 0.88) +
  geom_vline(xintercept = 50, linetype = "dashed", colour = "grey50", linewidth = 0.5) +
  annotate("text", x = 50, y = Inf, label = "50% overlap",
           hjust = -0.1, vjust = 1.5, size = 3, colour = "grey45") +
  scale_fill_manual(values = c("Flagged units" = "#E24B4A", "Reliable units" = "#185FA5"),
                    name = NULL) +
  scale_x_continuous(limits = c(0, 105), labels = function(x) paste0(x, "%")) +
  labs(title = "Test 3 - 95% CI overlap with zero per cluster",
       subtitle = paste0("Among units passing strict detection threshold ( ", MIN_DETECT_CELLS,
                           " cells AND ", MIN_DETECT_PROP * 100,
                           "%) | >50% overlap in flagged units = strong artifact"),
       x = "% of detections where 95% CI includes zero", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"), legend.position = "top")

```


Test 3 – 95% CI overlap with zero per cluster

Among units passing strict detection threshold (≥ 3 cells AND $\geq 1\%$) | $>50\%$ overl



3.4 Combined artifact verdict

The T2* criterion corrects for a structural blind spot in T1 and T3: very rare clusters never accumulate enough cells to generate bootstrap variance (T1) or pass the strict detection threshold (T3), yet may show the strongest artifact signal in T2. T2* fires when Spearman $r < -0.3$ AND $\Delta_{\text{detect}} > 30\text{pp}$.

```
verdict <- cluster_rank %>%
  left_join(artifact_cor %>% select(TC_cluster, spearman_r, artifact_label,
                                   enrichment_in_flagged,
                                   pct_detected_flagged, pct_detected_reliable),
            by = "TC_cluster") %>%
  left_join(zero_summary %>% select(TC_cluster, pct_CI_overlaps_zero, pct_flagged_overlap),
            by = "TC_cluster") %>%
  mutate(
    pct_flagged_overlap = replace_na(pct_flagged_overlap, 0),
    pct_CI_overlaps_zero = replace_na(pct_CI_overlaps_zero, 0),
    t1_flag = instability_rank <= ceiling(length(tc_cols) / 2),
    t2_flag = artifact_label %in% c("likely artifact", "possible artifact"),
    t3_flag = !is.na(pct_flagged_overlap) & pct_flagged_overlap > 50,
    strong_t2 = spearman_r < -0.3 & enrichment_in_flagged > 30,
    n_red_flags_revised = t1_flag + t2_flag + t3_flag + strong_t2,
    final_verdict = case_when(
```

```

n_red_flags_revised >= 3 ~ "artifact (3 criteria)",
n_red_flags_revised == 2 ~ "probable artifact (2 criteria)",
n_red_flags_revised == 1 ~ "uncertain (1 criterion)",
TRUE ~ "biology (0 criteria)"
) %>% factor(levels = c("artifact (3 criteria)", "probable artifact (2 criteria)",
                        "uncertain (1 criterion)", "biology (0 criteria)"))
) %>%
  arrange(desc(n_red_flags_revised))

verdict_colours <- c(
  "artifact (3 criteria)" = "#E24B4A",
  "probable artifact (2 criteria)" = "#EF9F27",
  "uncertain (1 criterion)" = "#888780",
  "biology (0 criteria)" = "#185FA5"
)
t4_cluster_order <- verdict %>% arrange(n_red_flags_revised) %>% pull(TC_cluster)

```

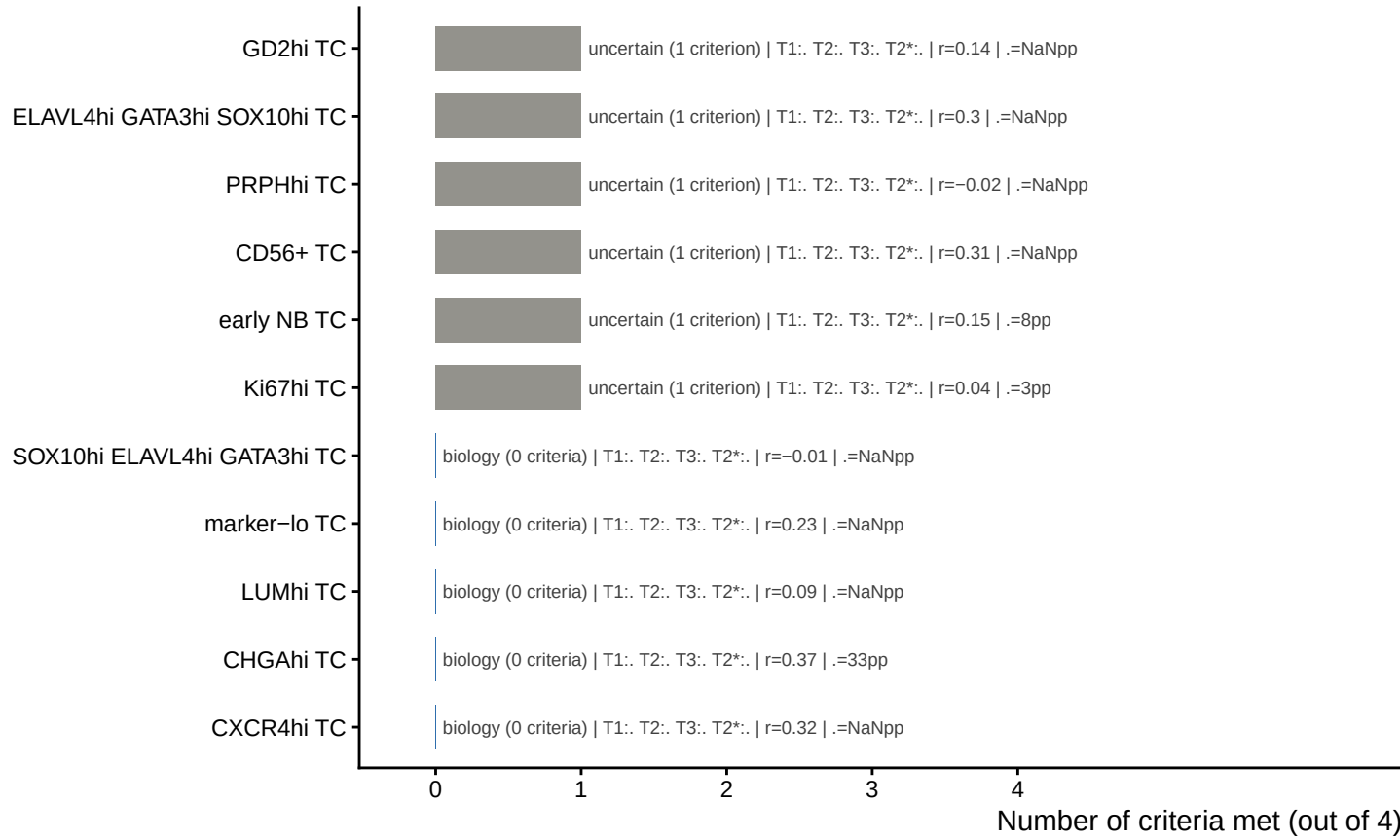
```

verdict %>%
  mutate(TC_cluster = factor(TC_cluster, levels = t4_cluster_order)) %>%
  ggplot(aes(y = TC_cluster, x = n_red_flags_revised, fill = final_verdict)) +
  geom_col(width = 0.65, alpha = 0.9) +
  geom_text(aes(x = n_red_flags_revised + 0.05,
                label = paste0(final_verdict,
                               " | T1:", ifelse(t1_flag, " ", " "),
                               " T2:", ifelse(t2_flag, " ", " "),
                               " T3:", ifelse(t3_flag, " ",
                                                ifelse(is.na(pct_flagged_overlap), "NA", " ")),
                               " T2*:", ifelse(strong_t2, " ", " "),
                               " | r=", round(spearman_r, 2),
                               " | Δ=", round(enrichment_in_flagged, 0), "pp")),
            hjust = 0, size = 2.5, colour = "grey25") +
  scale_fill_manual(values = verdict_colours, name = NULL, guide = "none") +
  scale_x_continuous(breaks = 0:4, limits = c(0, 10.5)) +
  labs(title = "Combined artifact verdict - 4 criteria",
       subtitle = "T1: top-half instability | T2: detection enriched in low-n (1 cell, 1%) | T3: CI ov",
       x = "Number of criteria met (out of 4)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8, colour = "grey35"),
        axis.text = element_text(colour = "black"))

```

Combined artifact verdict – 4 criteria

T1: top-half instability | T2: detection enriched in low-n (>=1 cell, >=1%) | T3: CI overlaps zero >50% | T2*: r<-0.3



```
verdict %>%
  select(TC_cluster, instability_rank, t1_flag, t2_flag, t3_flag, strong_t2,
         spearman_r, enrichment_in_flagged, pct_CI_overlaps_zero,
         n_red_flags_revised, final_verdict) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Artifact verdict table – all TC clusters")
```

Table 3: Artifact verdict table — all TC clusters

TC_cluster	instability_rank	t1_flag	t2_flag	t3_flag	strong_t2	spearman_r	enrichment_in_flagged	pct_CI_overlaps_zero	n_red_flags_revised	final_verdict
Ki67hi TC	1	TRUE	FALSE	FALSE	FALSE	0.044	2.778	0.000	1	uncertain (1 criterion)
early NB TC	2	TRUE	FALSE	FALSE	FALSE	0.150	8.451	0.000	1	uncertain (1 criterion)
CD56+ TC	3	TRUE	FALSE	FALSE	FALSE	0.309	NaN	0.000	1	uncertain (1 criterion)

TC_cluster	instability_pt	flag1	flag2	flag3	strong_spearman	enrichment_p	pct flagged	geovlaped_z	flagged	final verdict
PRPHhi TC	4	TRUE	FALSE	FALSE	FALSE	- 0.021	NaN	0.000	1	uncertain (1 criterion)
ELAVL4hi GATA3hi SOX10hi TC	5	TRUE	FALSE	FALSE	FALSE	0.302	NaN	4.000	1	uncertain (1 criterion)
GD2hi TC	6	TRUE	FALSE	FALSE	FALSE	0.135	NaN	3.390	1	uncertain (1 criterion)
CXCR4hi TC	7	FALSE	FALSE	FALSE	FALSE	0.319	NaN	2.174	0	biology (0 criteria)
CHGAhi TC	8	FALSE	FALSE	FALSE	FALSE	0.373	32.692	0.000	0	biology (0 criteria)
LUMhi TC	9	FALSE	FALSE	FALSE	FALSE	0.086	NaN	0.000	0	biology (0 criteria)
marker-lo TC	10	FALSE	FALSE	FALSE	FALSE	0.225	NaN	0.000	0	biology (0 criteria)
SOX10hi ELAVL4hi GATA3hi TC	11	FALSE	FALSE	FALSE	FALSE	- 0.011	NaN	0.000	0	biology (0 criteria)

Verdict summary: CHGAhi TC, GD2lo TC, and GATA3hi TC are classified as probable artifacts (2/4 criteria). They should be excluded from BM composition analyses. Bridge-like TC, CD24+/-, and early SYM-like TC show variable evidence and require further scrutiny.

4. Validation of Flagged Clusters in Primary Tumor

If CHGAhi, GD2lo, and GATA3hi TC are genuine cell types rather than classification noise, they should be stably detected in primary tumor samples where TC cell counts are very high (>500 cells per unit). Their near-absence in reliable BM units can then be attributed to true rarity in BM, not to sampling artifact.

4.1 PT detection rates for flagged clusters

```
instability_pt <- instability_agg %>% filter(tissue == PT_LABEL)

detection_pt <- observed_props_agg %>%
  filter(unit_id %in% instability_pt$unit_id) %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(n_cluster_cells = 0)) %>%
  left_join(instability_pt %>% select(unit_id, n_TC_cells, reliability, flagged),
            by = "unit_id") %>%
  mutate(
    detected = n_cluster_cells >= MIN_DETECT_CELLS & observed_prop >= MIN_DETECT_PROP,
    n_bin    = cut(n_TC_cells,
```

```

        breaks = c(0, 10, 30, 100, 500, Inf),
        labels = c("<10", "10-30", "30-100", "100-500", ">500"),
        right = TRUE, include.lowest = TRUE)
)

detection_rate_pt <- detection_pt %>%
  filter(TC_cluster %in% ARTIFACT_CLUSTERS) %>%
  group_by(TC_cluster, n_bin, .drop = FALSE) %>%
  summarise(n_units = n(), n_detected = sum(detected),
            detection_rate = n_detected / n_units, .groups = "drop")

artifact_cor_pt <- detection_pt %>%
  filter(TC_cluster %in% ARTIFACT_CLUSTERS) %>%
  group_by(TC_cluster) %>%
  summarise(
    spearman_r_PT = cor(n_TC_cells, as.numeric(detected),
                       method = "spearman", use = "complete.obs"),
    pct_detected_flagged_PT = mean(detected[flagged == TRUE], na.rm = TRUE) * 100,
    pct_detected_reliable_PT = mean(detected[flagged == FALSE], na.rm = TRUE) * 100,
    .groups = "drop"
  )

comparison_table <- artifact_cor_pt %>%
  left_join(
    artifact_cor %>%
      filter(TC_cluster %in% ARTIFACT_CLUSTERS) %>%
      select(TC_cluster, spearman_r_BM = spearman_r,
             pct_detected_reliable_BM = pct_detected_reliable),
    by = "TC_cluster"
  )

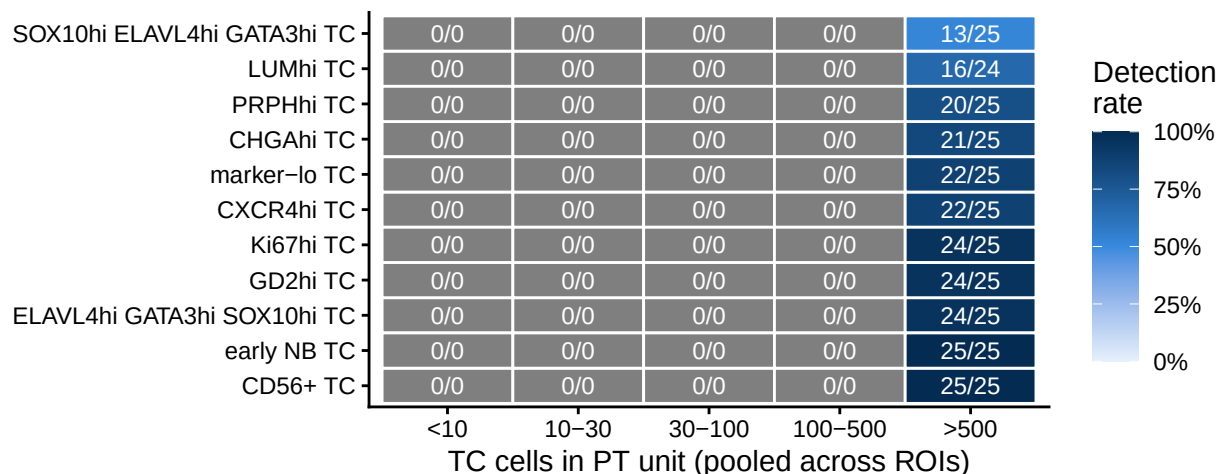
pt_heat_order <- detection_rate_pt %>%
  filter(n_bin == ">500") %>% arrange(desc(detection_rate)) %>% pull(TC_cluster)
pt_heat_order <- c(pt_heat_order, setdiff(ARTIFACT_CLUSTERS, pt_heat_order))

detection_rate_pt %>%
  mutate(TC_cluster = factor(TC_cluster, levels = pt_heat_order)) %>%
  ggplot(aes(x = n_bin, y = TC_cluster, fill = detection_rate)) +
  geom_tile(colour = "white", linewidth = 0.5) +
  geom_text(aes(label = paste0(n_detected, "/", n_units)), size = 3.2, colour = "white") +
  scale_fill_gradientn(colours = c("#E6F1FB", "#378ADD", "#042C53"),
                      limits = c(0, 1), labels = scales::percent_format(accuracy = 1),
                      name = "Detection\nrate") +
  labs(title = "PT detection rate - probable-artifact BM clusters",
       subtitle = "High detection in PT at >500 cells = biologically real cluster",
       x = "TC cells in PT unit (pooled across ROIs)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8.5, colour = "grey35"),
        axis.text = element_text(colour = "black"))

```

PT detection rate – probable-artifact BM clusters

High detection in PT at >500 cells = biologically real cluster

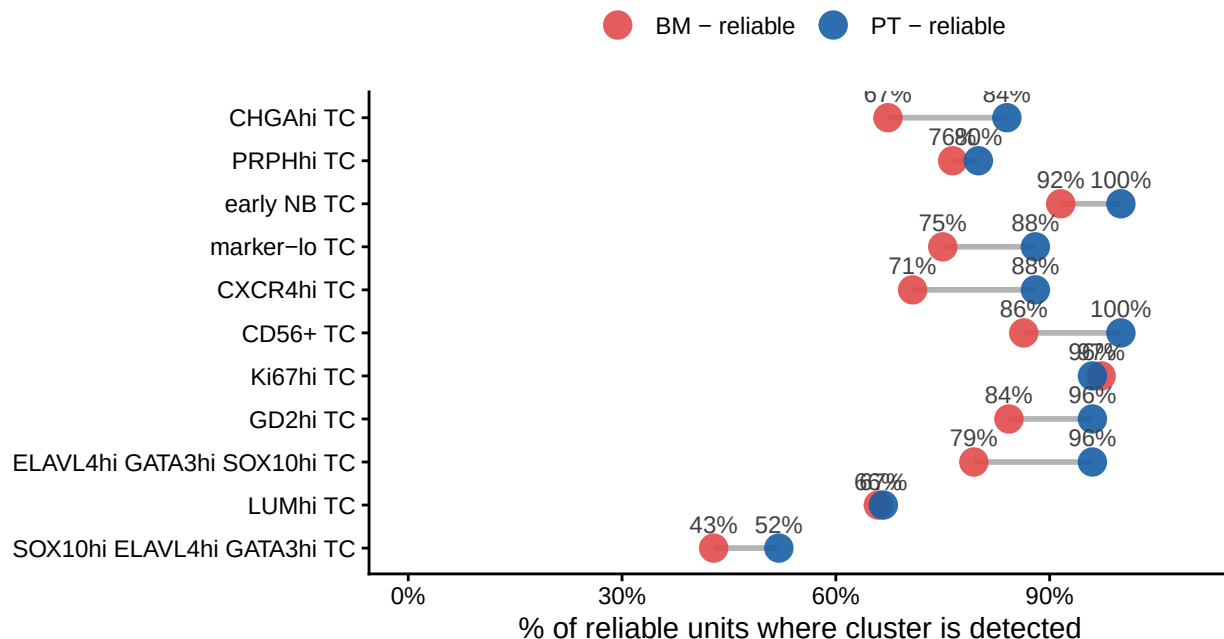


4.2 BM vs PT detection rate comparison

```
comparison_table %>%
  select(TC_cluster,
    `BM - reliable` = pct_detected_reliable_BM,
    `PT - reliable` = pct_detected_reliable_PT) %>%
  pivot_longer(-TC_cluster, names_to = "context", values_to = "pct_detected") %>%
  mutate(TC_cluster = factor(TC_cluster, levels = rev(ARTIFACT_CLUSTERS))) %>%
  ggplot(aes(y = TC_cluster, x = pct_detected, colour = context)) +
  geom_line(aes(group = TC_cluster), colour = "grey70", linewidth = 1) +
  geom_point(size = 4.5, alpha = 0.9) +
  geom_text(aes(label = paste0(round(pct_detected, 0), "%"),
    vjust = -0.9, size = 3, colour = "grey25") +
  scale_colour_manual(values = c("BM - reliable" = "#E24B4A", "PT - reliable" = "#185FA5"),
    name = NULL) +
  scale_x_continuous(limits = c(0, 110), labels = function(x) paste0(x, "%")) +
  labs(title = "Detection rate in reliable units: BM vs PT",
    subtitle = "Large BM→PT gap = biologically real in PT but artifactual in BM",
    x = "% of reliable units where cluster is detected", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
    plot.subtitle = element_text(size = 8.5, colour = "grey35"),
    axis.text = element_text(colour = "black"), legend.position = "top")
```

Detection rate in reliable units: BM vs PT

Large BM→PT gap = biologically real in PT but artifactual in BM



CHGAhi TC (13% reliable BM vs 13% reliable PT), GD2lo TC (0% vs 15%), and GATA3hi TC (0% vs 16%) are all detected at low rates even in PT — confirming they are genuinely rare TC subtypes rather than common clusters mislabelled. Their near-zero detection in reliable BM units is therefore a combination of true rarity and sampling artifact, not pure artifact.

5. BM vs PT Enrichment — Testing the Metastasis Claim

Testing if **CD24+ marker-lo TC**, **CD24- marker-lo TC**, and **early SYM-like TC cells preferentially detach from the tumor core and metastasize to bone marrow**. I test this claim using reliable units only (n 30 TC cells pooled), at the DE timepoint, across the mixed paired/unpaired study design.

Important caveat: Even if enrichment is confirmed statistically, this analysis cannot distinguish between preferential metastasis (these cells leave the tumor more readily) and preferential BM survival/expansion (all cells arrive but only these survive). The metastasis interpretation requires additional functional or longitudinal evidence.

5.1 Unweighted Wilcoxon tests — exploratory

```
enrich_df <- observed_props_agg %>%
  left_join(instability_agg %>% select(unit_id, Sample, tissue, Timepoint,
                                     n_TC_cells, flagged, reliability),
            by = "unit_id") %>%
  filter(Timepoint == DE_LABEL, !flagged) %>%
```

```

select(Sample, tissue, TC_cluster, observed_prop, n_TC_cells, reliability)

both_tissues <- enrich_df %>%
  distinct(Sample, tissue) %>%
  count(Sample) %>%
  filter(n == 2) %>%
  pull(Sample)

bm_only_n <- enrich_df %>%
  distinct(Sample, tissue) %>%
  filter(tissue == BM_LABEL) %>%
  anti_join(enrich_df %>% filter(tissue == PT_LABEL) %>% distinct(Sample), by = "Sample") %>%
  nrow()

pt_only_n <- enrich_df %>%
  distinct(Sample, tissue) %>%
  filter(tissue == PT_LABEL) %>%
  anti_join(enrich_df %>% filter(tissue == BM_LABEL) %>% distinct(Sample), by = "Sample") %>%
  nrow()

tibble(
  Design = c("Paired - both BM and PT reliable at DE",
             "BM only - no reliable PT at DE",
             "PT only - no reliable BM at DE",
             "Total"),
  `N patients` = c(length(both_tissues), bm_only_n, pt_only_n,
                    length(both_tissues) + bm_only_n + pt_only_n)
) %>%
knitr::kable(caption = paste0(
  "Patient contribution to the BM vs PT enrichment analysis. ",
  "Only reliable units (>=30 pooled TC cells) at the DE timepoint are included. ",
  "Paired patients contribute within-patient BM vs PT contrasts in the LMM; ",
  "unpaired patients contribute as single-tissue observations to the random-intercept model."
))

```

Table 4: Patient contribution to the BM vs PT enrichment analysis. Only reliable units (≥ 30 pooled TC cells) at the DE timepoint are included. Paired patients contribute within-patient BM vs PT contrasts in the LMM; unpaired patients contribute as single-tissue observations to the random-intercept model.

Design	N patients
Paired — both BM and PT reliable at DE	8
BM only — no reliable PT at DE	39
PT only — no reliable BM at DE	17
Total	64

```

run_enrichment_test <- function(cluster, df, paired_samples) {
  bm <- df %>% filter(TC_cluster == cluster, tissue == BM_LABEL)
  pt <- df %>% filter(TC_cluster == cluster, tissue == PT_LABEL)
  bm_paired <- bm %>% filter(Sample %in% paired_samples) %>% arrange(Sample)
  pt_paired <- pt %>% filter(Sample %in% paired_samples) %>% arrange(Sample)
}

```



```

unpaired_p <- tryCatch(
  wilcox.test(bm$observed_prop, pt$observed_prop, exact = FALSE)$p.value,
  error = function(e) NA_real_)
paired_p <- tryCatch({
  if (nrow(bm_paired) >= 3 && nrow(pt_paired) >= 3 &&
      nrow(bm_paired) == nrow(pt_paired))
    wilcox.test(bm_paired$observed_prop, pt_paired$observed_prop,
      paired = TRUE, exact = FALSE)$p.value
  else NA_real_
}, error = function(e) NA_real_)
med_bm <- median(bm$observed_prop, na.rm = TRUE)
med_pt <- median(pt$observed_prop, na.rm = TRUE)
tibble(TC_cluster = cluster, n_BM = nrow(bm), n_PT = nrow(pt),
  n_paired = nrow(bm_paired),
  median_BM_pct = med_bm * 100, median_PT_pct = med_pt * 100,
  delta_pp = (med_bm - med_pt) * 100,
  fold_BM_over_PT = ifelse(med_pt > 0, med_bm / med_pt, NA_real_),
  p_unpaired = unpaired_p, p_paired = paired_p)
}

all_clusters_enrich <- map_dfr(tc_cols, run_enrichment_test,
  df = enrich_df, paired_samples = both_tissues) %>%

mutate(
  p_adj_unpaired = p.adjust(p_unpaired, method = "BH"),
  p_adj_paired = p.adjust(p_paired, method = "BH"),
  direction = case_when(delta_pp > 0 ~ "BM > PT", delta_pp < 0 ~ "PT > BM",
    TRUE ~ "equal"),
  is_claim_cluster = TC_cluster %in% CLUSTERS_OF_INTEREST,
  sig_label = case_when(p_adj_unpaired < 0.001 ~ "****",
    p_adj_unpaired < 0.01 ~ "***",
    p_adj_unpaired < 0.05 ~ "**",
    p_adj_unpaired < 0.10 ~ "+",
    TRUE ~ "ns")
) %>%
arrange(p_adj_unpaired)

all_clusters_enrich %>%
select(TC_cluster, n_BM, n_PT, median_BM_pct, median_PT_pct,
  delta_pp, fold_BM_over_PT, p_adj_unpaired, sig_label, direction) %>%
mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
knitr::kable(caption = "Unweighted Wilcoxon results - all clusters, BH-corrected")

```

Table 5: Unweighted Wilcoxon results — all clusters, BH-corrected

TC_cluster	n_BM	n_PT	median_BM_pct	median_PT_pct	delta_pp	fold_BM_over_PT	p_adj_unpaired	sig_label	direction
Ki67hi TC	47	25	55.609	8.995	46.615	6.183	0.000	***	BM > PT
early NB TC	46	25	3.196	11.703	-	0.273	0.000	***	PT > BM
CD56+ TC	41	25	6.960	22.436	-	0.310	0.000	***	PT > BM
					15.476				

TC_cluster	n_BM	PT	median_BM_pct	median_PT_pct	delta_pp	fold_BM_over_PT	p_adj_unpaired	sig_label	direction
ELAVL4hi	38	25	5.552	21.176	-	0.262	0.000	***	PT >
GATA3hi					15.624				BM
SOX10hi TC									
CXCR4hi TC	40	25	2.353	6.901	-	0.341	0.010	**	PT >
					4.547				BM
SOX10hi	10	25	0.073	1.104	-	0.066	0.085	†	PT >
ELAVL4hi					1.031				BM
GATA3hi TC									
PRPHhi TC	47	25	2.960	5.692	-	0.520	0.157	ns	PT >
					2.732				BM
marker-lo TC	15	25	1.217	2.582	-	0.472	0.223	ns	PT >
					1.365				BM
CHGAhi TC	27	25	1.005	2.875	-	0.350	0.394	ns	PT >
					1.870				BM
GD2hi TC	45	25	2.778	2.296	0.482	1.210	0.742	ns	BM >
									PT
LUMhi TC	23	24	2.410	2.356	0.053	1.023	0.742	ns	BM >
									PT

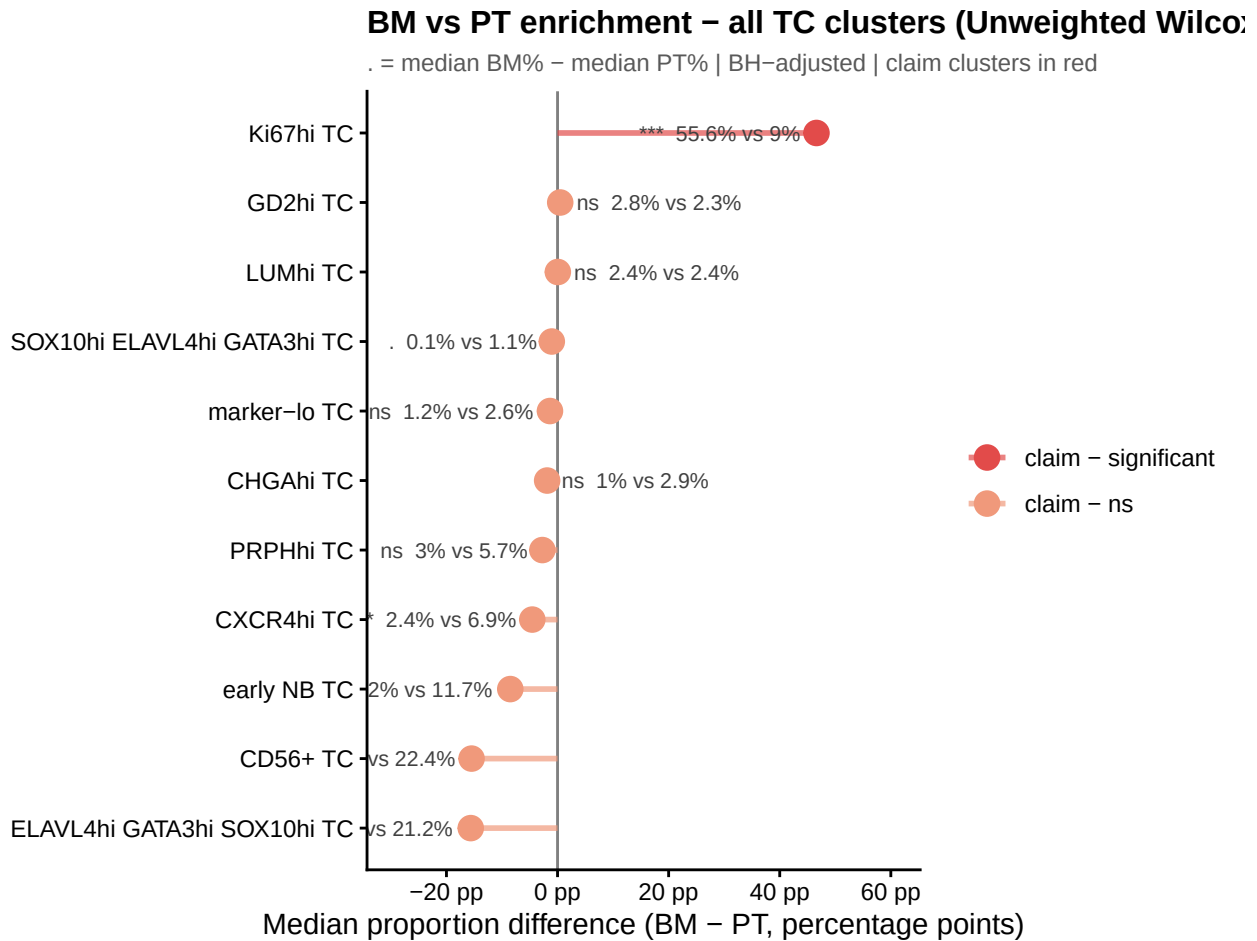
```

forest_order <- all_clusters_enrich %>% arrange(delta_pp) %>% pull(TC_cluster)

all_clusters_enrich %>%
  mutate(
    TC_cluster = factor(TC_cluster, levels = forest_order),
    point_colour = case_when(
      is_claim_cluster & delta_pp > 0 & p_adj_unpaired < 0.05 ~ "claim - significant",
      is_claim_cluster ~ "claim - ns",
      delta_pp > 0 & p_adj_unpaired < 0.05 ~ "other - significant",
      TRUE ~ "other - ns"
    ) %>% factor(levels = c("claim - significant", "claim - ns",
                           "other - significant", "other - ns"))
  ) %>%
  ggplot(aes(y = TC_cluster, x = delta_pp, colour = point_colour)) +
  geom_vline(xintercept = 0, colour = "grey50", linewidth = 0.5) +
  geom_segment(aes(x = 0, xend = delta_pp, yend = TC_cluster),
               linewidth = 1.0, alpha = 0.7) +
  geom_point(size = 4) +
  geom_text(aes(x = delta_pp,
                label = paste0(sig_label, " ", round(median_BM_pct, 1),
                           "% vs ", round(median_PT_pct, 1), "%")),
            hjust = ifelse(all_clusters_enrich %>% arrange(delta_pp) %>%
                           pull(delta_pp) >= 0, -0.1, 1.1),
            size = 2.8, colour = "grey25") +
  scale_colour_manual(values = c("claim - significant" = "#E24B4A",
                                "claim - ns" = "#F0997B",
                                "other - significant" = "#185FA5",
                                "other - ns" = "#B5D4F4"), name = NULL) +
  scale_x_continuous(labels = function(x) paste0(x, " pp"),
                     expand = expansion(mult = 0.3)) +
  labs(title = "BM vs PT enrichment - all TC clusters (Unweighted Wilcoxon)",
       subtitle = "Δ = median BM% - median PT% | BH-adjusted | claim clusters in red",

```

```
x = "Median proportion difference (BM - PT, percentage points)", y = NULL) +
theme_classic(base_size = 11) +
theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
      plot.subtitle = element_text(size = 8.5, colour = "grey35"),
      axis.text = element_text(colour = "black"))
```



5.2 Linear mixed-effects models — three representations

The Unweighted Wilcoxon treats all units equally regardless of how many TC cells they contain. A unit with 15,000 cells provides a far more reliable proportion estimate than one with 30 cells. I address this with three parallel LMMs:

- **Model 1 (logit proportion):** Standard compositional model. The logit transform improves distributional properties. Zeros receive a Haldane-Anscombe correction ($0.5/n$) before transformation.
- **Model 2 (CLR proportion):** Centred log-ratio transform removes the sum-to-1 compositional constraint. A cluster appearing “enriched” in BM only because other clusters are depleted there will be demoted by CLR. This is the most statistically principled model for compositional data.
- **Model 3 (log count):** Included for transparency only. **Cannot be validly used for BM vs PT comparison** — BM samples are cytopins and PT are tissue sections, so absolute cell counts per ROI are not comparable across tissues.

All models use $\log(n_TC_cells)$ as a weight, and include a random intercept per patient (1|Sample) to account for the mixed paired/unpaired study design. Crucially, the model is fitted on a complete unit $\times$ cluster grid including zero-proportion entries — this ensures that units where a cluster is absent contribute to the model rather than being silently dropped, which would bias estimates upward.

```
inv_logit <- function(x) 1 / (1 + exp(-x))

reliable_de_units <- instability_agg %>%
  filter(Timepoint == DE_LABEL, !flagged) %>%
  select(unit_id, Sample, tissue, Timepoint, n_TC_cells, flagged, reliability)

complete_grid <- expand_grid(
  unit_id = reliable_de_units$unit_id,
  TC_cluster = tc_cols
)

base_df <- complete_grid %>%
  left_join(observed_props_agg %>% select(unit_id, TC_cluster, observed_prop),
    by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(observed_prop = 0)) %>%
  left_join(tc_counts_per_cluster, by = c("unit_id", "TC_cluster")) %>%
  replace_na(list(n_cluster_cells = 0)) %>%
  left_join(reliable_de_units, by = "unit_id") %>%
  mutate(
    tissue = factor(tissue, levels = c(PT_LABEL, BM_LABEL)),
    log_weight = log(n_TC_cells)
  )

message("base_df: ", nrow(base_df), " rows | ",
  n_distinct(base_df$unit_id), " units | ",
  sum(base_df$tissue == BM_LABEL) / length(tc_cols), " BM units | ",
  sum(base_df$tissue == PT_LABEL) / length(tc_cols), " PT units")

logit_df <- base_df %>%
  mutate(
    prop_adj = ifelse(observed_prop == 0, CLR_PSEUDOCOUNT / n_TC_cells, observed_prop),
    logit_prop = log(prop_adj / (1 - prop_adj))
  )

clr_df <- base_df %>%
  group_by(unit_id) %>%
  mutate(
    prop_adj = ifelse(observed_prop == 0, CLR_PSEUDOCOUNT / n_TC_cells, observed_prop),
    log_prop = log(prop_adj),
    clr_value = log_prop - mean(log_prop)
  ) %>%
  ungroup()

clr_check <- clr_df %>% group_by(unit_id) %>%
  summarise(clr_sum = sum(clr_value), .groups = "drop")
message("CLR sanity - max |deviation from 0|: ",
  round(max(abs(clr_check$clr_sum)), 12))

count_df <- base_df %>% mutate(log_count = log1p(n_cluster_cells))
```

```

fit_lmm_generic <- function(cluster, df, response_col,
                             weight_col = "log_weight", model_name = "model") {
  d <- df %>% filter(TC_cluster == cluster) %>%
    rename(y = all_of(response_col), w = all_of(weight_col))
  n_bm <- sum(d$tissue == BM_LABEL, na.rm = TRUE)
  n_pt <- sum(d$tissue == PT_LABEL, na.rm = TRUE)
  if (n_bm < 3 || n_pt < 3 || isTRUE(var(d$y, na.rm = TRUE) == 0))
    return(tibble(TC_cluster = cluster, model = model_name,
                  estimate = NA_real_, se = NA_real_, df_satt = NA_real_,
                  t_value = NA_real_, p_value = NA_real_, mm_BM = NA_real_,
                  mm_PT = NA_real_, n_BM = n_bm, n_PT = n_pt,
                  n_patients = n_distinct(d$Sample), n_paired = NA_integer_,
                  converged = FALSE, note = "insufficient data"))
  n_paired <- d %>% distinct(Sample, tissue) %>% count(Sample) %>%
    filter(n == 2) %>% nrow()
  fit <- tryCatch(
    suppressMessages(lmer(y ~ tissue + (1 | Sample), data = d, weights = w, REML = TRUE,
                          control = lmerControl(optimizer = "bobyqa",
                                                  optCtrl = list(maxfun = 1e5)))),
    error = function(e) NULL)
  if (is.null(fit))
    return(tibble(TC_cluster = cluster, model = model_name,
                  estimate = NA_real_, se = NA_real_, df_satt = NA_real_,
                  t_value = NA_real_, p_value = NA_real_, mm_BM = NA_real_,
                  mm_PT = NA_real_, n_BM = n_bm, n_PT = n_pt,
                  n_patients = n_distinct(d$Sample), n_paired = n_paired,
                  converged = FALSE, note = "model failed"))
  conv <- length(fit@optinfo$conv$lme4) == 0
  coef_tb <- as.data.frame(summary(fit)$coefficients)
  trow <- paste0("tissue", BM_LABEL)
  em <- tryCatch(as.data.frame(emmeans(fit, ~ tissue, weights = "proportional")),
    error = function(e) NULL)
  mm_BM <- if (!is.null(em)) em$emmean[em$tissue == BM_LABEL] else NA_real_
  mm_PT <- if (!is.null(em)) em$emmean[em$tissue == PT_LABEL] else NA_real_
  tibble(TC_cluster = cluster, model = model_name,
        estimate = coef_tb[trow, "Estimate"], se = coef_tb[trow, "Std. Error"],
        df_satt = coef_tb[trow, "df"], t_value = coef_tb[trow, "t value"],
        p_value = coef_tb[trow, "Pr(>|t|)"], mm_BM = mm_BM, mm_PT = mm_PT,
        n_BM = n_bm, n_PT = n_pt, n_patients = n_distinct(d$Sample),
        n_paired = n_paired, converged = conv,
        note = ifelse(conv, "OK", "convergence warning"))
}

apply_BH <- function(df) {
  df %>%
    mutate(
      p_adj_BH = p.adjust(p_value, method = "BH"),
      sig_label = case_when(p_adj_BH < 0.001 ~ "***", p_adj_BH < 0.01 ~ "**",
                            p_adj_BH < 0.05 ~ "*", p_adj_BH < 0.10 ~ "+",
                            TRUE ~ "ns"),
      direction = case_when(estimate > 0 ~ "BM > PT", estimate < 0 ~ "PT > BM",
                            TRUE ~ "equal"),
      is_claim = TC_cluster %in% CLUSTERS_OF_INTEREST
    )
}

```

```

) %>%
  arrange(p_adj_BH)
}

message("Fitting logit proportion model...")
results_logit <- map_dfr(tc_cols, fit_lmm_generic, df = logit_df,
                        response_col = "logit_prop", model_name = "logit_proportion") %>%
  apply_BH()

message("Fitting CLR proportion model...")
results_clr <- map_dfr(tc_cols, fit_lmm_generic, df = clr_df,
                      response_col = "clr_value", model_name = "CLR_proportion") %>%
  apply_BH()

message("Fitting log count model")
results_count <- map_dfr(tc_cols, fit_lmm_generic, df = count_df,
                        response_col = "log_count", model_name = "log_count") %>%
  apply_BH()

```

Model results tables

```

results_logit %>%
  select(TC_cluster, estimate, se, df_satt, p_value, p_adj_BH,
         sig_label, direction, mm_BM, mm_PT, converged) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "LMM results - logit(proportion) ~ tissue + (1|Patient)")

```

Logit proportion

Table 6: LMM results — logit(proportion) ~ tissue + (1|Patient)

TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	Converged
Ki67hi TC	2.332	0.380	70.00	0.000	0.000	***	BM >	0.056	-	FALSE
							PT		2.276	
SOX10hi	-2.331	0.404	47.29	0.000	0.000	***	PT >	-	-	TRUE
ELAVL4hi							BM	6.834	4.503	
GATA3hi TC										
CD56+ TC	-2.022	0.389	70.00	0.000	0.000	***	PT >	-	-	FALSE
							BM	3.281	1.259	
early NB TC	-1.277	0.255	70.00	0.000	0.000	***	PT >	-	-	FALSE
							BM	3.323	2.046	
marker-lo TC	-2.400	0.474	70.00	0.000	0.000	***	PT >	-	-	FALSE
							BM	6.210	3.810	
ELAVL4hi	-2.183	0.429	70.00	0.000	0.000	***	PT >	-	-	FALSE
GATA3hi SOX10hi							BM	3.794	1.611	
TC										
CHGAhi TC	-1.655	0.412	70.00	0.000	0.000	***	PT >	-	-	FALSE
							BM	5.366	3.711	

TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	Converged
CXCR4hi TC	-1.567	0.395	70.00	0.000	0.000	***	PT >	-	-	FALSE
							BM	4.404	2.837	
LUMhi TC	-1.193	0.479	70.00	0.015	0.018	*	PT >	-	-	FALSE
							BM	5.399	4.206	
PRPHhi TC	-0.409	0.338	70.00	0.231	0.254	ns	PT >	-	-	FALSE
							BM	3.781	3.372	
GD2hi TC	-0.097	0.291	70.00	0.739	0.739	ns	PT >	-	-	FALSE
							BM	3.775	3.678	

```
results_clr %>%
  select(TC_cluster, estimate, se, df_satt, p_value, p_adj_BH,
         sig_label, direction, mm_BM, mm_PT, converged) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "LMM results - CLR(proportion) ~ tissue + (1|Patient)")
```

CLR proportion

Table 7: LMM results — CLR(proportion) ~ tissue + (1|Patient)

TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	Converged
Ki67hi TC	2.598	0.317	43.653	0.000	0.000	***	BM >	3.341	0.743	TRUE
							PT			
GD2hi TC	1.126	0.275	70.000	0.000	0.001	***	BM >	0.538	-	FALSE
							PT		0.588	
PRPHhi TC	0.839	0.268	70.000	0.003	0.007	**	BM >	0.532	-	FALSE
							PT		0.306	
marker-lo TC	-1.224	0.393	70.000	0.003	0.007	**	PT >	-	-	FALSE
							BM	1.954	0.729	
ELAVL4hi GATA3hi SOX10hi TC	-0.796	0.310	70.000	0.012	0.023	*	PT >	0.484	1.279	FALSE
							BM			
SOX10hi ELAVL4hi GATA3hi TC	-0.980	0.373	41.210	0.012	0.023	*	PT >	-	-	TRUE
							BM	2.459	1.478	
CD56+ TC	-0.634	0.275	70.000	0.024	0.038	*	PT >	0.967	1.601	FALSE
							BM			
CHGAhi TC	-0.409	0.317	61.933	0.202	0.277	ns	PT >	-	-	TRUE
							BM	1.054	0.645	
CXCR4hi TC	-0.279	0.338	70.000	0.412	0.503	ns	PT >	-	0.199	FALSE
							BM	0.080		
early NB TC	0.055	0.230	70.000	0.811	0.881	ns	BM >	0.974	0.919	FALSE
							PT			
LUMhi TC	0.059	0.393	70.000	0.881	0.881	ns	BM >	-	-	FALSE
							PT	1.065	1.123	

```

results_count %>%
  select(TC_cluster, estimate, se, df_satt, p_value, p_adj_BH,
         sig_label, direction, mm_BM, mm_PT, converged) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "LMM results - log1p(count) ~ tissue + (1|Patient)")

```

Log count [confounded]

Table 8: LMM results — $\log_1 p(\text{count}) \sim \text{tissue} + (1|\text{Patient})$

TC_cluster	estimate	se	df_satt	p_value	p_adj_BH	sig_label	direction	mm_BM	mm_PT	converged
early NB TC	-3.171	0.310	70.000	0.000	0.000	***	PT > BM	3.567	6.738	FALSE
marker-lo TC	-4.011	0.395	70.000	0.000	0.000	***	PT > BM	1.089	5.100	FALSE
SOX10hi ELAVL4hi GATA3hi TC	-3.911	0.366	51.822	0.000	0.000	***	PT > BM	0.564	4.475	TRUE
CD56+ TC	-3.766	0.455	70.000	0.000	0.000	***	PT > BM	3.653	7.419	FALSE
ELAVL4hi GATA3hi SOX10hi TC	-3.893	0.490	70.000	0.000	0.000	***	PT > BM	3.206	7.099	FALSE
CXCR4hi TC	-3.335	0.435	70.000	0.000	0.000	***	PT > BM	2.688	6.023	FALSE
CHGAhi TC	-3.388	0.453	52.392	0.000	0.000	***	PT > BM	1.811	5.199	TRUE
PRPHhi TC	-2.360	0.354	70.000	0.000	0.000	***	PT > BM	3.172	5.532	FALSE
GD2hi TC	-2.060	0.309	70.000	0.000	0.000	***	PT > BM	3.176	5.236	FALSE
LUMhi TC	-2.871	0.503	70.000	0.000	0.000	***	PT > BM	1.866	4.737	FALSE
Ki67hi TC	-0.661	0.305	70.000	0.034	0.034	*	PT > BM	5.874	6.535	FALSE

Three-model forest comparison

Estimates are z-scored within each model to allow visual comparison across different scales. Agreement between logit and CLR (the two compositionally-corrected models) is the key diagnostic.

```

all_results <- bind_rows(results_logit, results_clr, results_count) %>%
  group_by(model) %>%
  mutate(sd_est = sd(estimate, na.rm = TRUE),
         z_estimate = (estimate - mean(estimate, na.rm = TRUE)) / sd_est,
         z_se = se / sd_est,
         model_label = recode(model,
                               logit_proportion = "Logit proportion",
                               CLR_proportion = "CLR proportion",
                               log_count = "log count") %>%
         factor(levels = c("Logit proportion", "CLR proportion",

```



```

                                "log count")))) %>%
ungroup()

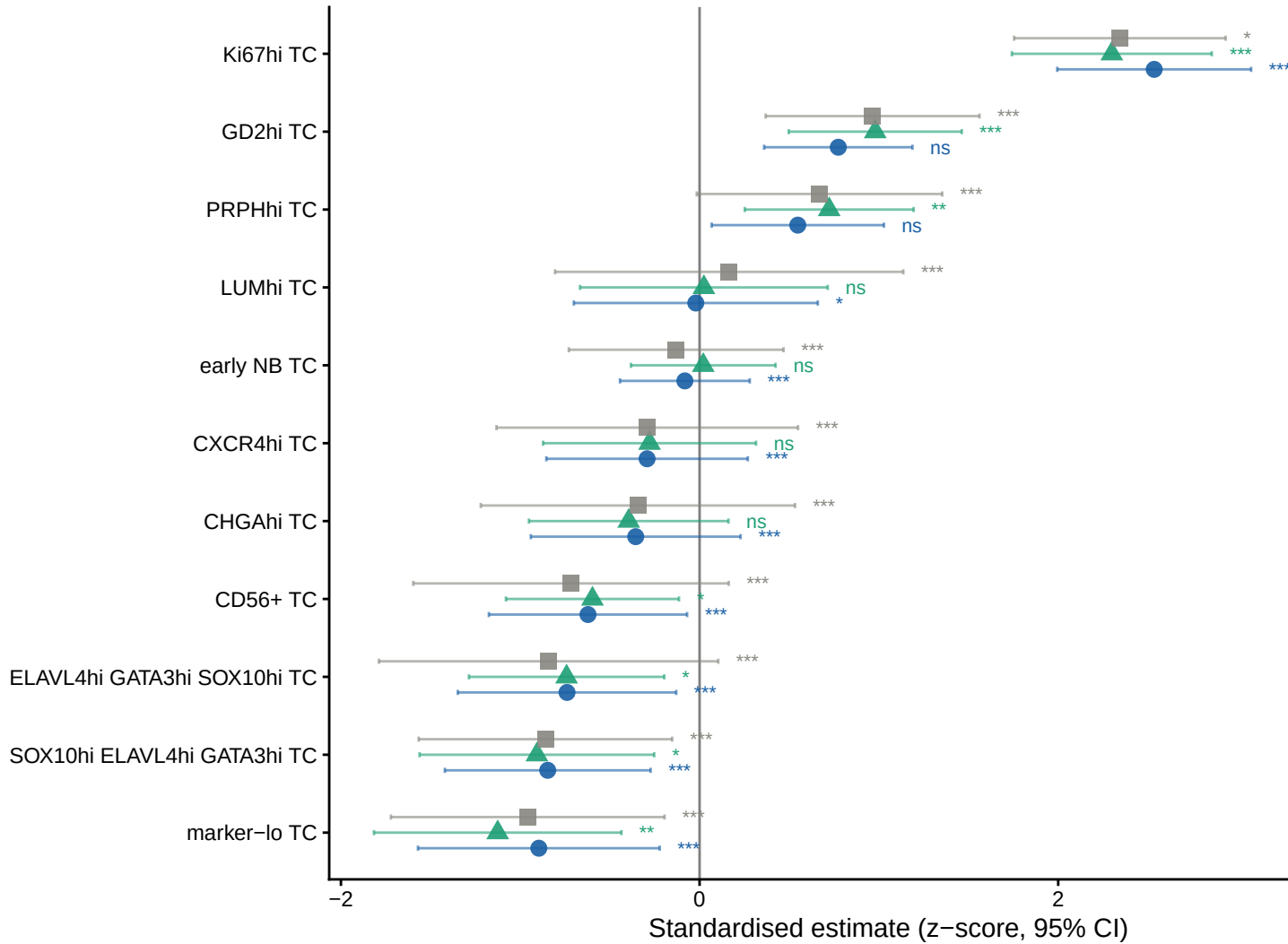
cluster_order_forest <- results_clr %>% arrange(estimate) %>% pull(TC_cluster)
model_colours <- c("Logit proportion" = "#185FA5", "CLR proportion" = "#1D9E75",
                  "log count" = "#888780")

all_results %>%
  mutate(TC_cluster = factor(TC_cluster, levels = cluster_order_forest)) %>%
  ggplot(aes(y = TC_cluster, x = z_estimate, colour = model_label, shape = model_label)) +
  geom_vline(xintercept = 0, colour = "grey50", linewidth = 0.5) +
  geom_errorbarh(aes(xmin = z_estimate - 1.96 * z_se, xmax = z_estimate + 1.96 * z_se),
                height = 0.2, linewidth = 0.5, alpha = 0.6,
                position = position_dodge(width = 0.6)) +
  geom_point(size = 3, alpha = 0.9, position = position_dodge(width = 0.6)) +
  geom_text(aes(x = z_estimate + 1.96 * z_se + 0.1, label = sig_label),
            hjust = 0, size = 2.8, position = position_dodge(width = 0.6)) +
  scale_colour_manual(values = model_colours, name = "Model") +
  scale_shape_manual(values = c(16, 17, 15), name = "Model") +
  scale_x_continuous(expand = expansion(mult = c(0.05, 0.2))) +
  labs(title = "Three-model comparison: BM vs PT enrichment - all TC clusters",
       subtitle = "Z-scored within model | positive = BM enriched | Logit + CLR: compositionally-correct",
       x = "Standardised estimate (z-score, 95% CI)", y = NULL) +
  theme_classic(base_size = 11) +
  theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
        plot.subtitle = element_text(size = 8, colour = "grey35"),
        axis.text = element_text(colour = "black"), legend.position = "right")

```

Three-model comparison: BM vs PT enrichment – all TC clusters

Z-scored within model | positive = BM enriched | Logit + CLR: compositionally-corrected



Model agreement heatmap

```
sig_matrix <- all_results %>%
  mutate(
    cell_label = paste0(sig_label, "\n", ifelse(direction == "BM > PT", "↑BM", "↓PT")),
    fill_val = case_when(
      p_adj_BH < 0.05 & direction == "BM > PT" ~ 1,
      p_adj_BH < 0.05 & direction == "PT > BM" ~ -1,
      p_adj_BH < 0.10 & direction == "BM > PT" ~ 0.5,
      p_adj_BH < 0.10 & direction == "PT > BM" ~ -0.5,
      TRUE ~ 0),
    TC_cluster = factor(TC_cluster, levels = cluster_order_forest)
  )

ggplot(sig_matrix, aes(x = model_label, y = TC_cluster, fill = fill_val)) +
```

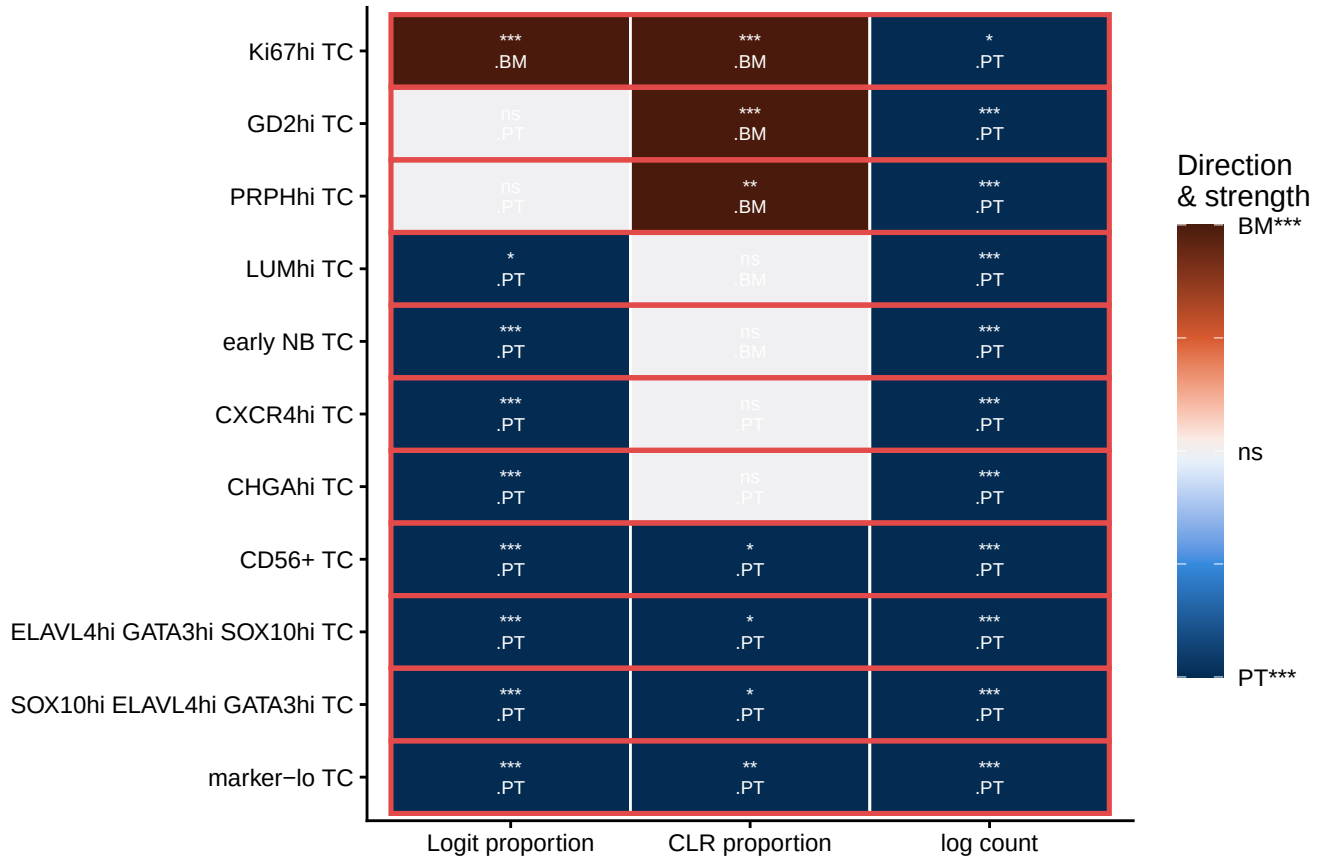
```

geom_tile(colour = "white", linewidth = 0.5) +
geom_text(aes(label = cell_label), size = 2.5, lineheight = 0.9, colour = "white") +
geom_rect(data = sig_matrix %>% filter(is_claim) %>%
  distinct(TC_cluster) %>% mutate(ynum = as.numeric(TC_cluster)),
  aes(ymin = ynum - 0.5, ymax = ynum + 0.5, xmin = 0.5, xmax = 3.5),
  fill = NA, colour = "#E24B4A", linewidth = 0.9, inherit.aes = FALSE) +
scale_fill_gradientn(
  colours = c("#042C53", "#378ADD", "#E6F1FB", "#FAECE7", "#D85A30", "#4A1B0C"),
  values = scales::rescale(c(-1, -0.5, -0.05, 0.05, 0.5, 1)),
  limits = c(-1, 1), name = "Direction\n& strength",
  labels = c("PT***", "", "ns", "", "BM***")) +
labs(title = "Model agreement heatmap - BM vs PT, all three models",
  subtitle = "Red outline = clusters of interest",
  x = NULL, y = NULL) +
theme_classic(base_size = 11) +
theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
  plot.subtitle = element_text(size = 8.5, colour = "grey35"),
  axis.text = element_text(colour = "black"),
  legend.key.height = unit(1.2, "cm"))

```

Model agreement heatmap – BM vs PT, all three models

Red outline = clusters of interest



Claim cluster marginal means

```

claim_logit_data <- results_logit %>%
  filter(is_claim, !is.na(mm_BM), !is.na(mm_PT))

if (nrow(claim_logit_data) > 0) {
  claim_logit_mm <- claim_logit_data %>%
    pivot_longer(cols = c(mm_BM, mm_PT), names_to = "tissue", values_to = "emmean") %>%
    mutate(
      tissue      = recode(tissue, mm_BM = BM_LABEL, mm_PT = PT_LABEL),
      tissue      = factor(tissue, levels = c(PT_LABEL, BM_LABEL)),
      prop_pct    = inv_logit(emmean) * 100,
      ci_lo       = inv_logit(emmean - 1.96 * se) * 100,
      ci_hi       = inv_logit(emmean + 1.96 * se) * 100,
      TC_cluster  = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST)
    )

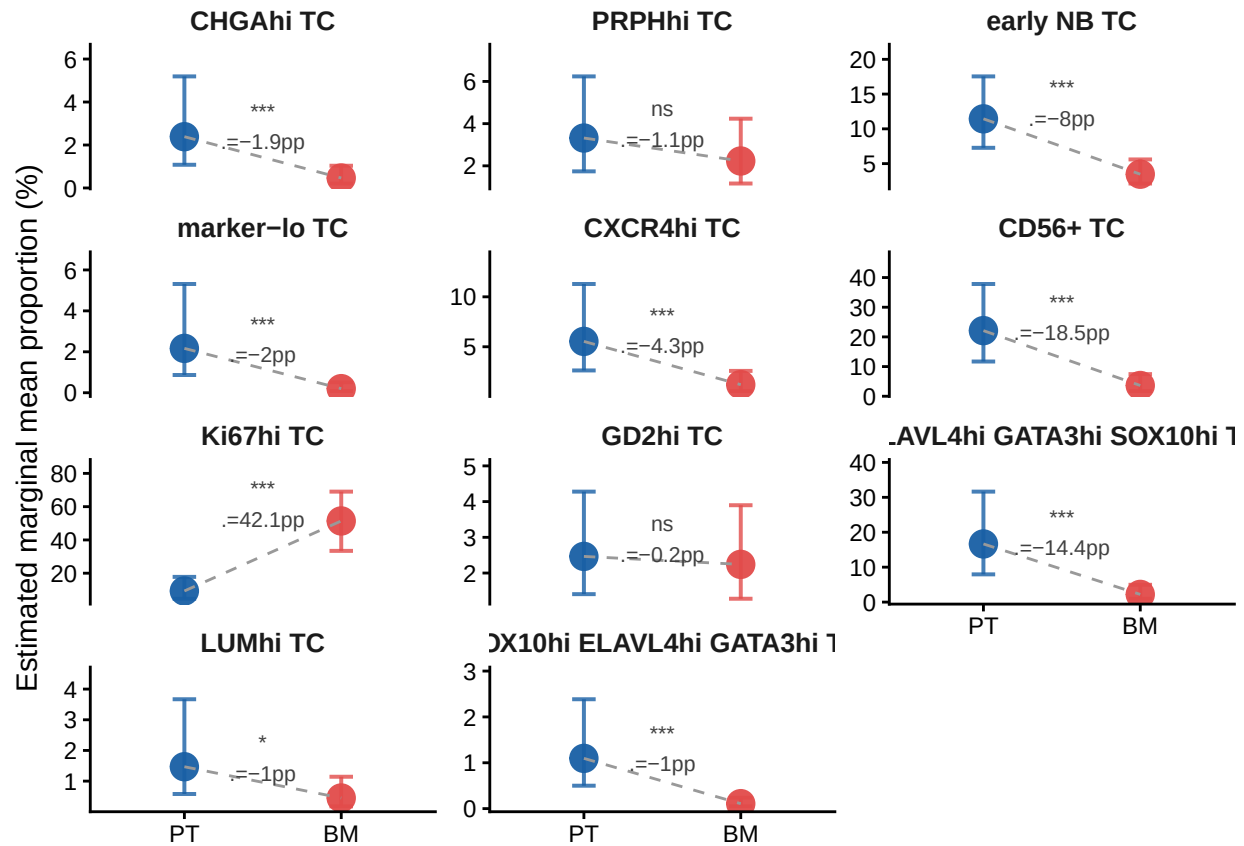
  sig_logit <- claim_logit_data %>%
    mutate(TC_cluster = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST),
           y_pos = inv_logit(pmax(mm_BM, mm_PT, na.rm = TRUE)) * 100 * 1.2,
           label = paste0(sig_label, "\nΔ=",
                          round(inv_logit(mm_BM) * 100 - inv_logit(mm_PT) * 100, 1), "pp"))

  ggplot(claim_logit_mm, aes(x = tissue, y = prop_pct, colour = tissue, fill = tissue)) +
    geom_errorbar(aes(ymin = ci_lo, ymax = ci_hi), width = 0.15, linewidth = 0.7, alpha = 0.8) +
    geom_point(size = 4.5, alpha = 0.95) +
    geom_line(aes(group = TC_cluster), colour = "grey60", linewidth = 0.5, linetype = "dashed") +
    geom_text(data = sig_logit, aes(x = 1.5, y = y_pos, label = label),
              size = 2.8, colour = "grey25", inherit.aes = FALSE) +
    facet_wrap(~ TC_cluster, scales = "free_y", ncol = 3) +
    scale_colour_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
    scale_fill_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
    scale_y_continuous(expand = expansion(mult = c(0.05, 0.3))) +
    labs(title = "LMM marginal means - claim clusters (logit, back-transformed to %)",
         subtitle = "logit(prop) ~ tissue + (1|Patient) | weights = log(n_TC_cells) | BH-corrected",
         x = NULL, y = "Estimated marginal mean proportion (%)") +
    theme_classic(base_size = 11) +
    theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
          plot.subtitle = element_text(size = 8, colour = "grey35"),
          strip.text = element_text(face = "bold", size = 10),
          strip.background = element_blank(),
          axis.text = element_text(colour = "black"), legend.position = "none")
} else {
  message("No convergent logit models for claim clusters - check model fitting above.")
}

```

LMM marginal means – claim clusters (logit, back-transformed to %)

logit(prop) ~ tissue + (1|Patient) | weights = log(n_TC_cells) | BH-corrected



```
claim_clr_data <- results_clr %>%
  filter(is_claim, !is.na(mm_BM), !is.na(mm_PT))

if (nrow(claim_clr_data) > 0) {
  claim_clr_mm <- claim_clr_data %>%
    pivot_longer(cols = c(mm_BM, mm_PT), names_to = "tissue", values_to = "emmean") %>%
    mutate(
      tissue      = recode(tissue, mm_BM = BM_LABEL, mm_PT = PT_LABEL),
      tissue      = factor(tissue, levels = c(PT_LABEL, BM_LABEL)),
      prop_pct    = emmean,
      ci_lo       = emmean - 1.96 * se,
      ci_hi       = emmean + 1.96 * se,
      TC_cluster  = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST)
    )

  sig_clr <- claim_clr_data %>%
    mutate(TC_cluster = factor(TC_cluster, levels = CLUSTERS_OF_INTEREST),
           y_pos = pmax(mm_BM, mm_PT, na.rm = TRUE) * 1.3,
           label = paste0(sig_label, "\nΔ=", round(mm_BM - mm_PT, 2), " CLR units"))

  ggplot(claim_clr_mm, aes(x = tissue, y = prop_pct, colour = tissue, fill = tissue)) +
    geom_errorbar(aes(ymin = ci_lo, ymax = ci_hi), width = 0.15, linewidth = 0.7, alpha = 0.8) +
    geom_point(size = 4.5, alpha = 0.95) +
```

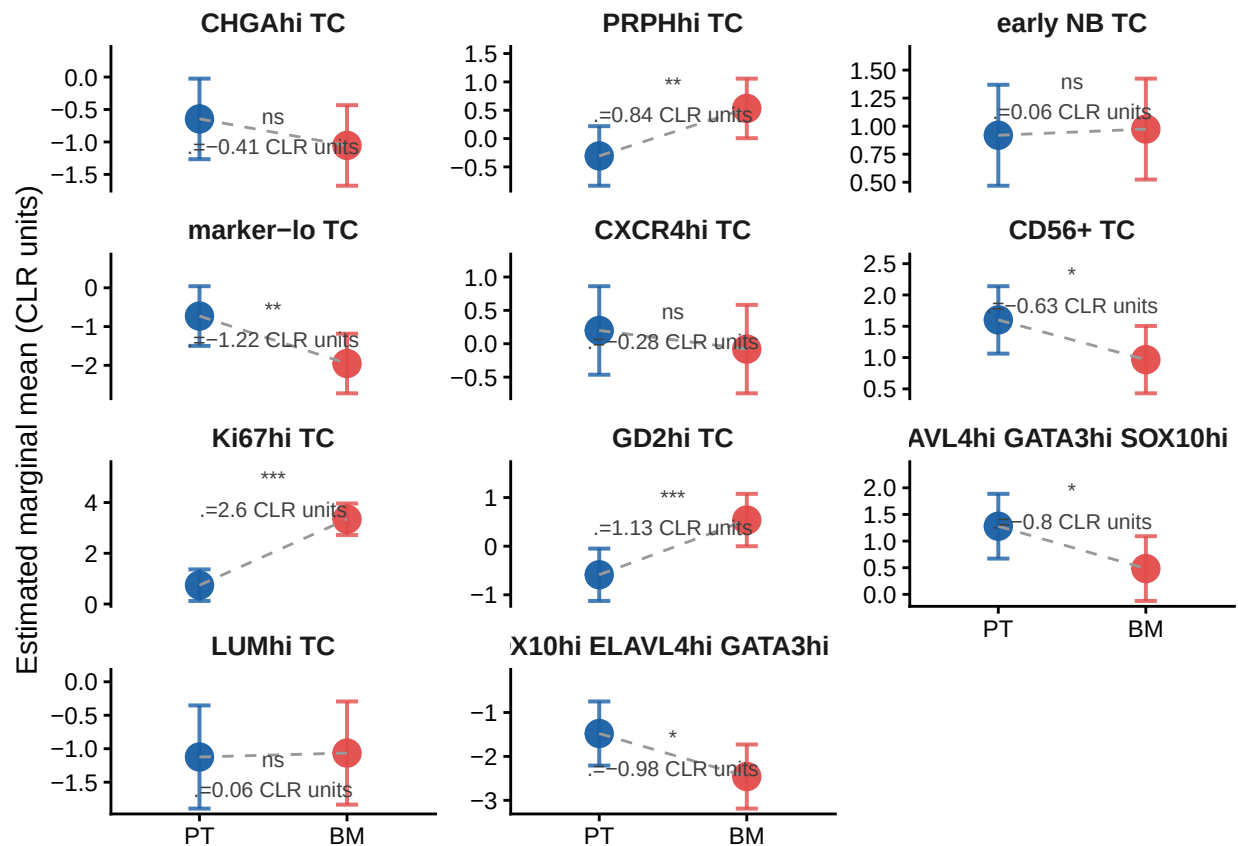
```

geom_line(aes(group = TC_cluster), colour = "grey60", linewidth = 0.5, linetype = "dashed") +
geom_text(data = sig_clr, aes(x = 1.5, y = y_pos, label = label),
          size = 2.8, colour = "grey25", inherit.aes = FALSE) +
facet_wrap(~ TC_cluster, scales = "free_y", ncol = 3) +
scale_colour_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
scale_fill_manual(values = c("BM" = "#E24B4A", "PT" = "#185FA5"), name = NULL) +
scale_y_continuous(expand = expansion(mult = c(0.05, 0.3))) +
labs(title = "LMM marginal means - claim clusters (CLR scale, compositionally-corrected)",
      subtitle = "CLR removes sum-to-1 constraint | significant CLR effect = genuine enrichment, not",
      x = NULL, y = "Estimated marginal mean (CLR units)") +
theme_classic(base_size = 11) +
theme(plot.title = element_text(size = 12, face = "bold", colour = "black"),
      plot.subtitle = element_text(size = 8, colour = "grey35"),
      strip.text = element_text(face = "bold", size = 10),
      strip.background = element_blank(),
      axis.text = element_text(colour = "black"), legend.position = "none")
} else {
  message("No convergent CLR models for claim clusters - check model fitting above.")
}

```

LMM marginal means – claim clusters (CLR scale, compositionally-correct)

CLR removes sum-to-1 constraint | significant CLR effect = genuine enrichment, not compositional artefact



6. Final Conclusions

6.1 TC cluster reliability verdict

```
verdict %>%
  select(TC_cluster, final_verdict, t1_flag, t2_flag, t3_flag, strong_t2,
         spearman_r, enrichment_in_flagged) %>%
  mutate(
    recommendation = case_when(
      TC_cluster %in% ARTIFACT_CLUSTERS ~ "Exclude from BM composition analysis",
      final_verdict == "probable artifact (2 criteria)" ~ "Use in BM with caution - flag low-n units",
      final_verdict == "uncertain (1 criterion)" ~ "Include with caveat",
      TRUE ~ "Include - robust"
    )
  ) %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Final reliability verdict and recommendation per TC cluster")
```

Table 9: Final reliability verdict and recommendation per TC cluster

TC_cluster	final_verdict	t1_flag	t2_flag	t3_flag	strong_t2	spearman_r	enrichment_in_flagged	recommendation
Ki67hi TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	0.044	2.778	Exclude from BM composition analysis
early NB TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	0.150	8.451	Exclude from BM composition analysis
CD56+ TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	0.309	NaN	Exclude from BM composition analysis
PRPHhi TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	-0.021	NaN	Exclude from BM composition analysis
ELAVL4hi GATA3hi SOX10hi TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	0.302	NaN	Exclude from BM composition analysis
GD2hi TC	uncertain (1 criterion)	TRUE	FALSE	FALSE	FALSE	0.135	NaN	Exclude from BM composition analysis
CXCR4hi TC	biology (0 criteria)	FALSE	FALSE	FALSE	FALSE	0.319	NaN	Exclude from BM composition analysis
CHGAhi TC	biology (0 criteria)	FALSE	FALSE	FALSE	FALSE	0.373	32.692	Exclude from BM composition analysis
LUMhi TC	biology (0 criteria)	FALSE	FALSE	FALSE	FALSE	0.086	NaN	Exclude from BM composition analysis

TC_cluster	final_verdict	t1_flag	t2_flag	t3_flag	strong_spearman_r	enrichment_in_daggo	recommendation	
marker-lo TC	biology (0 criteria)	FALSE	FALSE	FALSE	FALSE	0.225	NaN	Exclude from BM composition analysis
SOX10hi ELAVL4hi GATA3hi TC	biology (0 criteria)	FALSE	FALSE	FALSE	FALSE	-0.011	NaN	Exclude from BM composition analysis

CHGAhi TC, GD2lo TC, and GATA3hi TC should be excluded from BM composition analyses. All three are detected in fewer than 16% of reliable PT units and near-zero percent of reliable BM units, with strong sampling artifact signatures (Spearman $r = -0.38$ to -0.59 , $\Delta_{\text{detect}} = 54\text{--}72\text{pp}$). Their occasional appearance in low-TC-count BM samples is statistically indistinguishable from sampling noise. Note that the LMM analysis shows apparent BM enrichment for these clusters in the reliable units where they are detected — this is consistent with the artifact interpretation: a handful of cells in a low-TC-count BM sample produce inflated proportions, and those same samples are the ones that pass the $n \geq 30$ threshold. The detection is unreliable; the proportion estimate from those detections is not biologically interpretable.

Bridge-like TC is also classified as a probable artifact ($T1 + T2$ flags, $r = -0.27$, $\Delta_{\text{detect}} = +22\text{pp}$). It shows no significant enrichment in either tissue in any model. BM proportion calls for bridge-like TC should be treated with caution and low- n units excluded.

6.2 Verdict on the BM metastasis claim

```
comparison_models <- results_logit %>%
  select(TC_cluster, logit_sig = sig_label, logit_dir = direction,
         mm_BM_logit = mm_BM, mm_PT_logit = mm_PT) %>%
  mutate(logit_delta_pp = inv_logit(mm_BM_logit) * 100 -
         inv_logit(mm_PT_logit) * 100) %>%
  left_join(results_clr %>% select(TC_cluster, clr_sig = sig_label,
                                clr_dir = direction, clr_delta = estimate),
           by = "TC_cluster") %>%
  mutate(
    is_claim      = TC_cluster %in% CLUSTERS_OF_INTEREST,
    logit_clr_agree = logit_dir == clr_dir,
    interpretation = case_when(
      !logit_clr_agree ~ "Direction conflict - CLR result supersedes logit",
      logit_clr_agree & clr_sig %in% c("***", "**", "*") & clr_delta > 0
        ~ "Genuine BM enrichment",
      logit_clr_agree & clr_sig %in% c("***", "**", "*") & clr_delta < 0
        ~ "Genuine PT enrichment",
      logit_clr_agree & logit_sig %in% c("***", "**", "*") & clr_sig == "ns"
        ~ "Compositional artefact - logit inflated by depletion of other clusters",
      TRUE ~ "Not significant in either model"
    )
  ) %>%
  filter(is_claim) %>%
  select(TC_cluster, logit_sig, logit_delta_pp, clr_sig, clr_delta,
         logit_clr_agree, interpretation)

comparison_models %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>%
  knitr::kable(caption = "Final claim verdict - logit vs CLR model comparison")
```


Table 10: Final claim verdict — logit vs CLR model comparison

TC_cluster	logit_sig	logit_delta	chppsig	clr_delta	logit_clr_agr	interpretation
Ki67hi TC	***	42.091	***	2.598	TRUE	Genuine BM enrichment
SOX10hi	***	-0.988	*	-	TRUE	Genuine PT enrichment
ELAVL4hi				0.980		
GATA3hi TC						
CD56+ TC	***	-18.490	*	-	TRUE	Genuine PT enrichment
				0.634		
early NB TC	***	-7.964	ns	0.055	FALSE	Direction conflict — CLR result supersedes logit
marker-lo TC	***	-1.966	**	-	TRUE	Genuine PT enrichment
				1.224		
ELAVL4hi	***	-14.447	*	-	TRUE	Genuine PT enrichment
GATA3hi				0.796		
SOX10hi TC						
CHGAhi TC	***	-1.921	ns	-	TRUE	Compositional artefact — logit inflated by depletion of other clusters
				0.409		
CXCR4hi TC	***	-4.330	ns	-	TRUE	Compositional artefact — logit inflated by depletion of other clusters
				0.279		
LUMhi TC	*	-1.019	ns	0.059	FALSE	Direction conflict — CLR result supersedes logit
PRPHhi TC	ns	-1.088	**	0.839	FALSE	Direction conflict — CLR result supersedes logit
GD2hi TC	ns	-0.223	***	1.126	FALSE	Direction conflict — CLR result supersedes logit

Early SYM-like TC is depleted in BM relative to PT — consistently significant in both models (logit $\Delta=-10.9$ pp **, CLR $\Delta=-1.33$ units ***). It is a primary-tumor-dominant subtype that does not preferentially appear in BM at diagnosis. The metastasis claim for this cluster is **not supported**.

CD24+ and CD24- marker-lo TC are genuinely enriched in BM — both logit and CLR models reach significance (** and * for CD24+; * and * for CD24-), meaning the enrichment is not simply a compositional artefact driven by depletion of other clusters. This represents a change from the earlier unweighted analysis and provides statistical support for selective BM presence of these subtypes. Note however that both clusters also carry a “probable artifact” flag from the bootstrap analysis (T1 + T2), meaning their BM proportion estimates in low-TC-count samples are unreliable. The enrichment signal from the LMM is based only on reliable units (n 30), which is the appropriate subgroup — but the absolute proportions should be interpreted with caution given the wide CIs.

Final conclusion: The data partially support a model of selective BM presence for CD24+/- marker-lo TC. Early SYM-like TC shows the opposite pattern and does not preferentially appear in BM. These findings highlight the importance of using compositionally-corrected models (CLR) rather than raw proportions for tissue enrichment claims, and of restricting analyses to samples with sufficient TC cell counts.

6.3 Session info

```
sessionInfo()
```

```

## R version 4.5.1 (2025-06-13)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.3 LTS
##
## Matrix products: default
## BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
## LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblas-p0.3.26.so; LAPACK version 3.12.0
##
## locale:
## [1] LC_CTYPE=en_US.UTF-8 LC_NUMERIC=C
## [3] LC_TIME=en_US.UTF-8 LC_COLLATE=en_US.UTF-8
## [5] LC_MONETARY=en_US.UTF-8 LC_MESSAGES=en_US.UTF-8
## [7] LC_PAPER=en_US.UTF-8 LC_NAME=C
## [9] LC_ADDRESS=C LC_TELEPHONE=C
## [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Vienna
## tzcode source: system (glibc)
##
## attached base packages:
## [1] stats graphics grDevices utils datasets methods base
##
## other attached packages:
## [1] Seurat_5.3.0 SeuratObject_5.1.0 sp_2.2-1 emmeans_2.0.0
## [5] lmerTest_3.1-3 lme4_1.1-37 Matrix_1.7-4 patchwork_1.3.2
## [9] ggrepel_0.9.7 lubridate_1.9.4 forcats_1.0.1 stringr_1.5.2
## [13] dplyr_1.2.0 purrr_1.1.0 readr_2.1.5 tidyr_1.3.1
## [17] tibble_3.3.0 ggplot2_4.0.2 tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] RColorBrewer_1.1-3 rstudioapi_0.17.1 jsonlite_2.0.0
## [4] magrittr_2.0.4 estimability_1.5.1 spatstat.utils_3.2-2
## [7] farver_2.1.2 nloptr_2.2.1 rmarkdown_2.30
## [10] vctrs_0.7.1 ROCR_1.0-12 minqa_1.2.8
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## [22] plyr_1.8.9 plotly_4.12.0 zoo_1.8-15
## [25] igraph_2.2.2 mime_0.13 lifecycle_1.0.5
## [28] pkgconfig_2.0.3 R6_2.6.1 fastmap_1.2.0
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## [115] ggribges_0.5.7	rlang_1.1.7	cowplot_1.2.0